

MASTER

The use of (cognitive) strategies and technological tools to manage everyday life with Long COVID

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The use of (cognitive) strategies and technological tools to manage everyday life with Long COVID

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1375946

in partial fulfilment of the requirements for the degree of

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in Human-Technology Interaction**

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Abstract

The urgency is high to develop management methods for people with Long COVID so they can control their illness better and resume (part of) their lives. However, research on self-management methods is lacking, especially regarding cognition-involved activities. This thesis aims to contribute to the visibility of Long COVID, gain insight into its impact on everyday lives by giving people with Long COVID a voice, and develop knowledge and ideas on supporting and managing the cognitive aspects of everyday life. This was done with a human-centered, qualitative approach through a human-technology interaction lens.

This (explorative) thesis investigated the (potential) role of technology in the management of cognitive aspects of the everyday life of people with Long COVID. Using a participant-friendly research approach, including a sociodemographic questionnaire, an activity diary, and an in-depth interview, we gained insights from eleven people with Long COVID experiencing impaired cognitive functioning into the impact on everyday life, the (cognitive) strategies and (technological) tools they use, and the gaps that remain in existing support systems. The results showed that the current role of technology is mainly regarding complaint management, while the potential is mostly in supporting information processing within activities. For future technology design, tools should be developed that address the needs and desires of people with Long COVID. Based on these and other findings, this thesis concludes with methodological, research, and technology design recommendations.

Keywords: Long COVID, cognitive strategies, technological tools, information-processing, self-management, human-technology interaction

Table of Contents

Abstract.....	1
1. Introduction.....	5
2. Theoretical background.....	7
2.1 Long COVID	7
2.1.1 Manifestation of Long COVID	7
2.1.2 Group of interest	8
2.2 Self-management.....	10
2.2.1 Self-management methods.....	10
2.2.2 Self-management in the Long COVID context	11
2.2.3 Potential of (assistive) technology for people with Long COVID.....	13
2.3 Rationale study	15
2.3.1 Methodology literature	16
3. Methodology.....	18
3.1 Design	18
3.2 Participants	18
3.3 Setting.....	19
3.4 Materials.....	20
3.4.1 Sociodemographic questionnaire.....	20
3.4.2 Activity diary.....	20
3.4.3 Interview	21
3.5 Procedure	22
3.6 Data analysis	24
3.6.1 Context participants	24
3.6.2 Activity diaries	24
3.6.3 Thematic analysis	24
4. Results	25
4.1 Observations from the activity diary	25
4.2 Thematic analysis.....	26
4.2.1 Experiences with the activity diary	27
4.2.2 Theme 1 – Prevent everyday life	28
4.2.3 Theme 2 – Elusive nature	29

4.2.4	Theme 3 – Social environment.....	30
4.2.5	Theme 4 – Self-image.....	31
4.2.6	Theme 5 – Mental coping.....	33
4.2.7	Theme 6 - Trade-off desires and needs.....	33
4.2.8	Theme 7 – Everyday planning and activities	35
4.2.9	Theme 8 – (Technological) tools.....	36
4.2.10	Theme 9 – Desires technology.....	37
5.	Discussion.....	38
5.1	Managing uncertainties: People with Long COVID are expert control engineers of their everyday life.....	38
5.2	Extending cognition: Facilitating cognitive processes through active implementation of strategies and tools.....	39
5.3	High potential, but the realization of cognitive support through technological tools is missing	40
5.4	Interpretation-based conclusions.....	41
5.4.1	Impact on everyday life	41
5.4.2	Self-management of Long COVID	41
5.4.3	The potential of technology	42
5.5	Recommendations for future work	43
5.5.1	Methodological limitations and recommendations	43
5.5.2	Recommendations for research and design.....	44
6.	Conclusion	47
7.	Acknowledgements	48
	References.....	49
	Appendix A: Literature	59
	Appendix A.1: Symptom list Davis	59
	Symptom categories.....	59
	Symptom list.....	59
	Appendix B: Methodology	62
	Appendix B.1: Positionality statement	62
	Position regarding the group of interest	62
	Sociodemographic background	62
	Reflexivity during the data collection and analysis	63
	Appendix B.2: Descriptions of the participants	65
	Appendix C: Materials.....	69

Appendix C.1: Recruitment material	69
Recruitment text	69
Recruitment flyer.....	70
Appendix C.2: Briefing and debriefing.....	71
Briefing – phone	71
Debriefing after interview	73
Appendix C.3: Consent form.....	74
Appendix C.4: Sociodemographic questionnaire.....	79
Appendix C.5: Activity diary.....	82
Appendix C.5.1: Activity diary guidelines	82
Appendix C.5.2: Activity diary form.....	91
Appendix C.6: Interview schedule	92
Briefing interview	92
Interviewvragen	92
Debriefing interview en onderzoek.....	94
Appendix E: Data analysis	95
Appendix E.1: Whiteboard codes and themes	95
Appendix E.2: Original interview quotes	96
Appendix F: Flyer conclusions for participants	99

1. Introduction

During and after the coronavirus pandemic, it appeared that 10-20% of the people who got infected with the COVID-19 virus developed long-term complaints (World Health Organization, 2022). This is called Long COVID, and worldwide, at least 65 million people have or had this condition, a number that is still increasing¹ (Davis et al., 2023). A new health pandemic is born.

The daily lives of people with Long COVID are impacted because of impaired physical, cognitive, and/or social functions (Brus et al., 2024). Although most people infected with the coronavirus are only affected a little and recover within months, there remains a group of people who develop severe and systematic symptoms for a much longer period of time (O'Connor et al., 2024; Office for National Statistics, 2023). Their abilities to e.g., work, go to school, socialize, care for themselves, and do physical exercise are affected severely (Brus et al., 2024). This group of people needs to actively adjust their lives to survive the day and execute everyday tasks.

Currently, in the Netherlands, approximately 90,000 people have severe Long COVID (Dankers et al., 2023; Ministerie van Sociale Zaken en Werkgelegenheid, 2023; Office for National Statistics, 2023)² and even more people are limited in their regular daily living (having milder Long COVID than the so-called 'severe cases'). This has substantial consequences, like worsened mental health (Atchison et al., 2023; Hossain et al., 2023; Medinger & Altmann, 2022) and the development of neurological disorders (Everaerd & Biere-Rafi, 2023; Lopez-Leon et al., 2021; Taquet et al., 2021). Thus, the impact of Long COVID on the quality of life is enormous for many people.

This, of course, also impacts society; Long COVID leads to high societal costs as a consequence of decrease in productivity, incapacity to work, costs of care (not only acute care but also long-term care like home care), loss of children and young people in education, decreasing participation of people in other areas like informal care (Brus et al., 2024; Maatschappelijk Impact Team, 2023a). Also, Long COVID and the further development of new cases will lead to substantial societal costs, and an increasing group of people who are medically unfit for work (Maatschappelijk Impact Team, 2023b).

Despite the far-reaching impact, there is no proven cure against Long COVID yet (Bonilla et al., 2023; Fox et al., 2023). Acknowledgement of the existence of Long COVID has taken quite some time because of several reasons including the medical assumptions on the course (of virus) conditions, lack of educational materials for health care professionals, and the available health care resources (Katz et al., 2023; Moreels et al., 2024; Turk et al., 2024). As a result, the awareness of long-term complaints caused by a coronavirus infection has been lacking among health care professionals, politicians, citizens, etc. An example of this is the awareness of the manifestation of oral complaints among dental professionals (Varghese, 2024). A substantial consequence of this is this lack of

¹ It is important to take into account that this thesis was written in the academic year of 2024-2025. The numbers that are used are published before September 2024, meaning that when this article is read, the numbers and literature findings can be outdated.

² The estimation of how many people in the Dutch population suffer from severe Long COVID complaints is based on the proportion of people who answered "Yes, a lot" to the question "Does this (long covid, red.) reduce your ability to carry out day-to-day activities compared with the time before you had COVID- 19?" in the survey that was executed among the population of the United Kingdom by their Office for National Statistics (2023) (Maatschappelijk Impact Team, 2023b).

attention is that the funds for research on this topic were and are still relatively small regarding the needs (Callard & Perego, 2021). Besides, the condition has only existed for a short time and is seen as complex, which complicates the initiative of executing research (Medinger & Altmann, 2022; Perumal et al., 2023). Thus, relevant research for a remedy is still young, meaning that only limited insights have been gathered through studies.

At this moment in time, self-management methods are seen as the most effective way to treat and live with Long COVID (Dankers et al., 2023; Gloeckl et al., 2023; Ladds et al., 2024). These methods should enable people with Long COVID to execute everyday tasks and improve cognitive functions, so they can get back to regular daily living and participate in society again (Dankers et al., 2023). Besides that these methods could serve as treatment, they can also be valuable for the large group of people who do not recover by supporting them in their everyday life, e.g., in activities they are not able to do because of their impairments (Brus et al., 2024). More and more research on self-management methods and assistive tools is being done. However, these studies mostly include physical-focused treatment interventions or technologies. This is a limited view of how one can manage and implement strategies in one's everyday life; daily activities include a lot of cognitive functioning. As Long COVID often impairs the (executive) cognitive functions, it is valuable to look deeper into how these can be supported (Davis et al., 2021).

This thesis aims to contribute to more insight into how people with life-limiting Long COVID can be supported in the cognitive aspects of their everyday lives using a human-centered, qualitative approach focusing on the use of technology as assistance. Therefore, this study investigates the current role of technology in facilitating cognitive functioning in everyday life with Long COVID and explores how technology can be designed or adapted to improve this cognitive support in the future. The corresponding research question is *“What is the (potential) role of technology in the management of cognitive aspects of everyday life of people with Long COVID?”*

To answer the main research question, the impact of Long COVID on everyday lives and the implementation of cognitive strategies and supporting technologies need to be established first. Therefore, the following sub-research questions are answered first: RQ1: *“How does Long COVID affect the everyday lives of people who experience fatigue, cognitive dysfunction, and eventual Post Exertional Malaise-related symptoms that reduce their ability to carry out day-to-day activities severely?”* and RQ2: *“How do people with Long COVID-related fatigue and cognitive dysfunction adjust and implement cognitive strategies and technology in order to manage the cognitive aspects their everyday life?”*. Additionally, the potential and ideas of the lives experience experts need to be identified as well, which is done by answering the following question: RQ3: *“What do people with Long COVID currently miss in their developed strategies and implemented technological supporting tools to carry-out daily cognition-involved activities and what can be improved?”*.

2. Theoretical background

2.1 Long COVID

Long COVID³ is an illness that can develop after a SARS-CoV-2 (COVID-19) infection, leading to impairments in physical, cognitive, and/or social functioning. There are multiple definitions and names to describe this condition (Munblit et al., 2022; National Academies of Sciences, Engineering, and Medicine, 2024). This thesis follows the definition and notation of the World Health Organization (WHO) that is adopted by many research projects and governmental institutions in, among others, the Netherlands and the European Union. The WHO defines Long COVID as “the continuation or development of new symptoms 3 months after the initial SARS-CoV-2 infection, with these symptoms lasting for at least 2 months with no other explanation” (World Health Organization, 2022).

Long COVID can affect all infected people regardless of age, health, disability, socioeconomic status, gender, ethnicity, and living situation. These factors can however influence the severity and manifestation of the condition (Atchison et al., 2023; National Academies of Sciences, Engineering, and Medicine, 2024). To understand the impact of Long COVID on people’s lives, it is important to get some more insight into its manifestation and its complex workings.

2.1.1 Manifestation of Long COVID

Long COVID can manifest in single or multiple symptoms and single or multiple diagnosable conditions (National Academies of Sciences, Engineering, and Medicine, 2024). A prominent study led by people with Long COVID showed that there are more than 200 symptoms that people can experience long after their COVID-19 infection, with fatigue, cognitive dysfunction, and Post-Exertional Malaise (PEM) as the most common ones (Davis et al., 2021). The symptoms can be categorized into ten categories, like systemic, musculoskeletal, and neuropsychiatric symptoms (Davis et al., 2021). For further information and understanding, you can find the list of Davis et al. (2021) with symptoms categorized by type in [Appendix A.1: Symptom list Davis](#).

Long COVID is seen as a complex condition that is difficult to recover from, as multiple (organ) systems within the body are involved (Medinger & Altmann, 2022; Perumal et al., 2023). In practice, this leads to individuals who often suffer from multiple symptoms in variable compositions with differing severity and duration, leading to divergent disease profiles (Medinger & Altmann, 2022). For example, while some individuals lose their sense of smell and taste for a few months, others suffer from long-term systemic fatigue, severe brain fog, and worsening symptoms after (minimal) exertion for years. Another consequence of this is that symptoms are often closely connected, which makes it even more complicated for people with Long COVID to execute everyday tasks.

Although the exact underlying systems are difficult to identify, it seems that Long COVID shares similar grounds with other conditions regarding mechanisms and symptoms. Long COVID is associated with other (chronic) conditions like ME/CFS (Myalgic Encephalomyelitis/Chronic Fatigue

³ Because of the short existence of the disease, research about this condition is currently in full development. As a result, the research community has yet to agree upon many important things like underlying mechanisms, characteristics, and manifestations of the condition. In this thesis, we had to make certain choices regarding e.g., definitions. As the field is short-existing and there is a lack of consensus, the choices are not always well-substantiated but based on what is commonly used in for example the Netherlands, or European-based academic papers.

Syndrome), and fibromyalgia (Medinger & Altmann, 2022). Besides, people with Long COVID can develop other conditions like diabetes or dysautonomia (Appelman et al, 2024; Dorrnoro-Zubiete et al, 2024; National Academies of Sciences, Engineering, and Medicine et al., 2024). These conditions are also seen as complex; for many of them, no effective treatment has been found yet. Because of the similar nature, this study's methodological approach and findings could be beneficial for insights about the management of these related (semi-)chronic conditions as well.

2.1.2 Group of interest

In this thesis, the focus will be on people with Long COVID who are less capable of executing day-to-day tasks than before they had Long COVID, and experience persistent symptoms. This includes people with severe Long COVID, but also people who have less severe Long COVID but are still considerably limited in their everyday lives. This group of people needs to make adjustments in their lives to survive the day and execute (cognitive) daily tasks.

As this thesis investigates supporting cognitive functioning in everyday life, it is important to understand the symptoms that actively contribute to impaired cognitive (executive) functioning.

Cognitive dysfunction

First of all, the main symptom of difficulties in executing cognition-involved tasks; cognitive dysfunction. Cognitive dysfunction due to Long COVID affects operational processes and abilities of the brain. Zhao et al. (2024) indeed observed that a direct (objective) consequence of cognitive dysfunction is a lower cognitive processing speed. However, the exact manifestation of cognitive dysfunction due to Long COVID occurs in different combinations. In the leading symptoms list of Davis et al. (2021), cognitive complaints are mainly embodied by 1) brain fog, 2) memory issues, and 3) speech and language issues.

Brain fog is characterized by confusion, short-term memory loss, dizziness, and inability to concentrate, leading to a feeling of cloudiness in one's cognitive awareness (Garg et al., 2021). Further, memory issues can manifest in short- and long-term memory loss, but also in, for example, the inability to make new memories. Speech and language problems, on the other hand, occur through difficulties finding the right words, communicating verbally, processing text, etc. (Davis et al., 2021). As a result, cognitive dysfunction affects the ability to execute activities in everyday life; daily tasks like work, decision-making, or cooking get complicated and sometimes even impossible to perform for people with Long COVID (Davis et al., 2021).

Fatigue

Cognitive dysfunction and its manifestation in everyday life are highly associated with fatigue (Davis et al., 2021; Gloeckl et al., 2023). Fatigue in Long COVID is a different experience than 'usual' tiredness; people with Long COVID characterize this fatigue as a systemic exhaustion in every physical and cognitive part of the body (Medinger & Altmann, 2022). In conversations with people with Long COVID by journalist Ed Yong, the fatigue was described as completely draining, which results in substantial life changes influencing people's capabilities, social contact, and (mental) well-being (Yong, 2023). Many people with Long COVID are permanently tired. This means that they wake up fatigued and get more exhausted when their energy battery becomes emptier during the day. Fatigue has a major impact on cognitive functioning. Therefore, it is seen as the leading symptom for difficulties in cognition-involved everyday tasks, together with cognitive dysfunction.

Post-Exertional Malaise

Lastly, Post-Exertional Malaise (PEM) is also a common symptom that burdens everyday life, and on the other hand, is associated with the severity of symptoms and (cognitive) functioning (Appelman et al., 2024; Davis et al., 2021). It is important to note that it is not always associated with cognitive dysfunction, but it has a massive impact on how people manage and deal with their Long COVID and severely affects the ability to execute everyday tasks. Therefore, it will be actively considered in this study and the accompanying method.

PEM means that after excessive exercise that does not fit within the limited exercise capacity of their bodies, symptoms worsen or relapse. As a result, people with Long COVID who have PEM-related complaints should monitor the boundaries of their condition so their illness will not become worse (Appelman et al., 2024; Davis et al., 2021). Triggers could be cognitive exercise, but also physical and social activities; everything that takes effort is a trigger. PEM complicates the rehabilitation and recovery process and limits the lives of people with Long COVID even more. PEM is also often associated with ME/CFS (Gloeckl et al., 2023). An example of how PEM works is that cognitive fatigue relates to physical fatigue and pain, meaning that one can feel physically tired or develop pain in their muscles after cognitive effort. You could say that PEM is the worst-case manifestation of the underlying connections and involvement of multiple (organ) systems in the body; it amplifies the complexity of living with Long COVID.

Taken together, people with Long COVID who suffer from fatigue, cognitive dysfunction, and/or PEM need to do a cost-benefit analysis for every activity. Even the simplest day-to-day tasks like showering or cooking can be a great burden as they include physical strength but also ask for cognitive resources to execute all the involved steps. In Figure 1, you find a diagram that shows the underlying relations between the different symptoms.

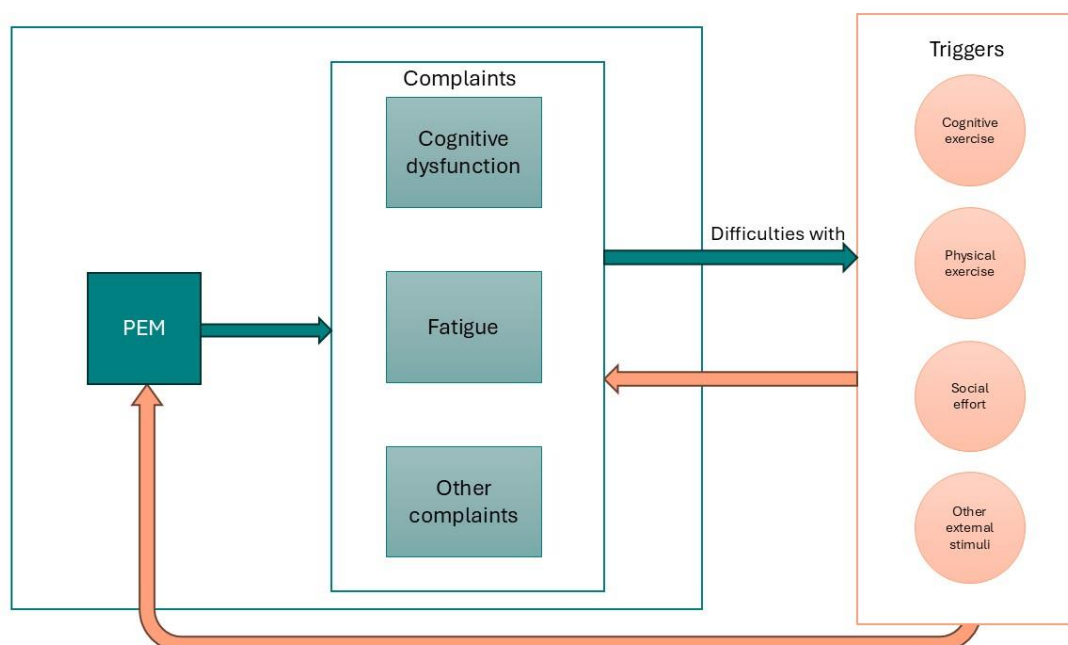


Figure 1: Diagram presenting the underlying connection of triggers and Long COVID symptoms

2.2 Self-management

To get insight into how technology currently and potentially plays a role in managing the cognitive aspects of everyday life, it is important to get a better understanding of current findings on how one can manage health-related complaints in everyday life. Therefore, we will look into the literature about self-management methods, how this is applied in the context of Long COVID, and the implementation of technology and other cognitive artifacts to manage complaints and cognitive activities.

2.2.1 Self-management methods

Currently, there is no proven, working medical remedy against Long COVID. Although there is a wide range of treatment methods for Long COVID, most of them were tested among a too small or uncontrolled sample. Besides, they apply different underlying pathophysiologic bases (Bonilla et al., 2023; Fox et al., 2023). Therefore, at the moment, self-management methods seem to be the most effective; several researchers emphasize that self-management is the most effective way to treat Long-COVID (Dankers et al., 2023; Gloeckl et al., 2023; Ladds et al., 2024). The World Health Organization also advises this to people with Long COVID; self-management strategies such as pacing, and planning rest moments in the day are recommended as a Long COVID fatigue rehabilitation intervention (World Health Organization, 2021, 2023).

Self-management techniques are also effectively used in other chronic conditions⁴ like ME/CFS and diabetes, and help people manage their condition (Barlow et al., 2002; Newman et al., 2004). Self-management is defined by Barlow et al. (2002) in their leading paper on the management of chronic conditions in everyday life, as “the individual’s ability to manage symptoms, treatment, physical, and psychosocial consequences and lifestyle changes inherent in living with a chronic condition”. They say that to be effective, people need to be able to understand the manifestation of their condition and apply cognitive, behavioral, and emotional measures to keep a decent quality of life. This means that one needs to be actively involved in the development of their symptom and controlling the adjustments of their lives and the (psychosocial) impact on their life (Newman et al., 2004). Self-management methods can be applied through, e.g., (cognitive) strategies, adjustments in life standards, and the use of (cognitive) artifacts.

Cognitive strategies and artifacts

People with Long COVID who experience fatigue and cognitive dysfunction (and potential PEM-related symptoms) benefit from self-management methods that focus on reducing (cognitive) load and executing day-to-day activities that involve cognitive functioning. This can be done by managing so-called ‘cognitive strategies’. Cognitive strategies can be defined in several ways depending on the research area, e.g., cognitive behavioral therapy or education (Nowakowski & Bieling, 2022; Pressley & Levin, 2012). In this thesis, cognitive strategies are defined as mental routines or procedures

⁴ Although a major part of people with Long COVID recover within a few months, some people suffer from long-lasting complaints for years and are not sure to get better. Although there is hope that they recover from Long COVID, they have many things in common with people with a chronic condition regarding taking into consideration their complaints in everyday life. Besides, Long COVID is associated with several chronic conditions like ME/CFS and diabetes. Therefore, severe (or at least limiting), persistent Long COVID can be seen as a (semi-)chronic condition.

people consciously use to perform a certain (cognitive) task, following the description of Dijk & Kintsch (1983).

In case of cognitive dysfunction because of Long COVID, the cognitive strategies people used before their life-changing SARS-CoV-2 infection do not fit their situation anymore; their cognitive capabilities are affected, and so is their cognitive functioning. This means that if they want to perform certain activities that involve cognitive processing, their cognitive strategies need to be adjusted so they suit their current cognitive functioning. This can be done by restructuring thinking and reorganizing cognitive processes. Through changing the procedure to execute a certain task or through extending the procedure through the use of cognitive artifacts⁵, like assistive technology tools, the cognitive functions are complemented such that people can perform (cognitive) activities that they usually cannot do or do not do well (Heersmink, 2015).

Epistemic actions

A common controlled cognitive strategy is the use of epistemic actions. For people with Long COVID, epistemic actions could play a key role in coping with their brain condition in combination with stimuli and normative standards for daily cognitive functioning. Epistemic actions rearrange the components of the world that are involved in the action in such a way that the cognitive process is adjusted to a more manageable procedure that is easier to execute (Kirsch & Maglio, 1994). This simplification leads to a reduced cognitive load, which positively contributes to the cognitive capability of a person (e.g., through reduced involved memory, number of steps, and chance of error) (Clark & Chalmers, 1998; Kirsch & Maglio, 1994). The adjustments can take place within the internal mind, but also by reorganizing the living and/or task environment, or extending cognitive possibilities with cognitive artifacts.

Through the implementation of cognitive artifacts like assistive technology, cognitive processes can be simplified or at least become more manageable using (artificial) epistemic actions. This can lead to 1) either a possibility to perform an action that otherwise would have been difficult or impossible; or 2) it reduces mental workload, meaning there is more (cognitive) capacity to perform other, upcoming tasks. As a result, cognitive artifacts like assistive technologies can enhance the cognitive functioning of people with Long COVID; utilizing epistemic actions, the cognitive functioning capacity is extended, which could also be beneficial for future activities.

2.2.2 Self-management in the Long COVID context

Despite the short existence of Long COVID, there have already been developed instructions for self-management strategies. In a review paper on rehabilitation methods for Long COVID, it was emphasized that cognitive dysfunction should be treated by focusing on developing strategies to rebuild everyday life and improving attention, memory, and executive functions (Gloeckl et al., 2023). This finding is supported by Dutch governmental organizations, for example, in a compilation of knowledge on Long COVID by the Dutch institution for medicine use (Instituut Verantwoord Medicijngebruik) (Dankers et al., 2023).

⁵ Artifacts are objects that are created by or modified through humans. These serve as tool and are developed with a specific purpose in mind (Meriam-Webster, 2024; Preston, 2022). Cognitive artifacts are tools that support the process of certain cognitive tasks (Carroll & Campbell, 1989).

Another common finding is the use of so-called ‘pacing strategies’, which are essential to control PEM-related complaints regarding everyday activities. Pacing teaches people how to manage their condition regarding activities without crashing; by means of experiencing and monitoring symptoms, people develop consciousness of to what extent activities are suitable for them and recognize the need for rest to prevent themselves from worsening their symptoms (Gloeckl et al., 2023; Medinger & Altmann, 2022; Stapleton et al., 2024).

Lastly, an insightful study by Leggat et al. (2024) on self-management to cope with Long COVID shows that people with Long COVID often develop their own strategies to manage their lives. These methods often focus on self-awareness through monitoring, trial-and-error, seeking knowledge, building in pleasure and comfort, and prioritizing the needs of the individual (Leggat et al., 2024).

It is important to note that self-management methods for Long COVID can be used in rehabilitation and coping with everyday life. On the one hand, strategies can be used to improve symptoms and rebuild one’s life (in case recovery is possible). On the other hand, self-management methods can enable people to deal with their condition regarding day-to-day activities through, e.g., pacing technologies and extending capabilities using epistemic actions.

Shortcomings of current self-management intervention research

Although this thesis focuses on the implementation of technology to manage the cognitive symptoms in executing daily tasks, it is valuable to get some more insight into existing studies about (non-technology-driven) methods that are currently used to manage Long COVID regarding everyday life. These have several substantial shortcomings, and by acknowledging these, this thesis can prevent them from happening.

Existing research on (self-management) interventions for Long COVID focuses mainly on the effect of the methods on specific symptoms, while this condition is seen as complex because of the involvement of multiple (organ) systems within the body. Training interventions often only include energy or fatigue management training (Rybkina et al., 2024; Stapleton et al., 2024; Weise et al., 2024). So, these only focus on the symptoms of fatigue, meaning they disregard the complexity and interdependence of Long COVID manifestations.

Besides, these intervention studies often only investigate the impact of the execution or use of these interventions on people’s lives, while the exact implementation of (cognitive) strategies and tools like technological devices in daily living processes and environment is missing (Dorronzoro-Zubiete et al., 2024; Stapleton et al., 2024; Weise et al., 2024). Knowledge and insight on the actual implementation of cognitive artifacts are essential for further development of interventions, optimization of cognitive strategies, and insight into how cognitive artifacts like technological tools can play a role in the management of the everyday lives and rehabilitation of people with Long COVID. Therefore, this thesis focuses on how people with Long COVID implement (cognitive) strategies and artifacts.

2.2.3 Potential of (assistive) technology for people with Long COVID

This thesis takes a human-technology interaction focus on the cognitive self-management of everyday activities for people with Long COVID. To better understand the (potential) role of technology to support the everyday life of people with Long COVID, it is valuable to look into how people with conditions that result in similar accessibility needs to Long COVID use technology and collect existing knowledge on technology use by people with Long COVID themselves⁶.

We can make three observations from (in)formal sources regarding technology use to support the everyday life of people with Long COVID. 1) Technology use among people with Long COVID-related chronic conditions shows the potential of technology as a helpful tool in everyday life; 2) There are only a few technological tools dedicated to Long COVID in particular; and 3) Existing technologies are used by people with Long COVID to support self-management.

1) Technology use among people with similar accessibility needs shows the potential.

Taking a human-technology interaction focus on the topic of supporting everyday life with Long COVID is a logical step; implementing technology in everyday lives to extend (cognitive) capacities in case of impaired characteristics also happens in case of other (chronic) conditions. We can observe this in the knowledge and experiences of other groups with similar accessibility needs and (cognitive) capacity challenges; many people deviate from the societal or their individual norm because of a (physical, cognitive, or mental) condition on management of daily life could be insightful for people with Long COVID as well. In the management of other conditions, it is seen that (assistive) technology often plays a role in enabling day-to-day activities (Bigham & Carrington, 2018; Howard et al., 2022; McGowan & McGregor, 2023).

Related work shows the potential of technology in the management of Long COVID in everyday lives; there are multiple examples of academic papers towards the use of technology in other (semi-) chronic conditions like fibromyalgia, chronic fatigue disease, and ADHD that involve Long COVID-related symptoms. Examples of self-management technologies used in chronic conditions are self-tracking in Parkinson's disease, migraine, and Multiple Sclerosis (Ayobi, 2020; Mishra et al., 2019; Schroeder et al., 2018), apps to monitor and manage mood for people who suffer from depression- or anxiety-related symptoms (Orcha Health, 2021), and a digital rehabilitation program for people with chronic obstructive pulmonary disease (Wienroth et al., 2020).

2) There are only a few technological tools dedicated to Long COVID in particular.

It is seen that there are only a few technological interventions developed for Long COVID specifically⁷. It is observed that the short existence, relative unfamiliarity, misunderstanding, absence of money, and complexity of the disease result in a lack of research on the condition (Alwan, 2022;

⁶ As there is only little academical research on human-technology interaction in Long COVID management, informal sources are considered; these still give relevant insight into the use of technological tools to support people with Long COVID. The sources in this section are examples of technologies, journalistic magazines, television programs, newspapers, and articles from specialist journals or websites, or semi-academic research organizations.

⁷ After thorough research on Google (Scholar), the researchers only found a few technological solutions that support people with Long COVID regarding the management of cognitive aspects of everyday life. This means that in the meantime, when you read this, probably more tools have been developed for Long COVID specifically.

Callard & Perego, 2021; Medinger & Altmann, 2022; Perumal et al., 2023). You could argue that this reality translates into the development of valid tools for Long COVID; this is lacking as well.

Although there are not many Long COVID-specific tools, it should be noted that initiatives are rising. There are two examples of Long COVID-specific apps that help you manage your Long COVID symptoms and thus your everyday life. The (most commonly used and known) apps *Fimo Health* and *Visible* record data on the symptoms of their users, which is needed to support them in everyday life by extending knowledge on their condition, and establishing strategies that fit their symptoms (Fimo Health, 2024; VisibleHealth, 2024). These examples indicate the potential of Long COVID-specific technology that enables people to execute everyday activities.

3) Existing technologies are used by people with Long COVID to support self-management.

On the other hand, existing technologies that are designed for another user group than people with Long COVID are often used to manage Long COVID symptoms regarding everyday life (Khondakar & Kaushik, 2023; Orcha Health, 2021). This is identified by AbilityNet. This charity organization that aims to make digital technology accessible for everyone, despite age or disability, provided a list of existing (not specifically for people with Long COVID) technologies that can support people with their Long COVID symptoms. They divide the tools by symptom type, resulting in technologies to help people with fatigue- and cognition-related symptoms, joint pain, and anxiety and depression (AbilityNet, 2024). Examples of these technological supporting tools are time management apps, text-to-speech software, and mood monitoring tools (AbilityNet, 2024; Orcha Health, 2021).

Another type of technology that was developed with another purpose, but which is valuable for self-management of Long COVID, are wearable devices like smart rings that measure biometrics (Khondakar & Kaushik, 2023). Although these are often used without any adaptations to their original form, they are also used by extending them with Long COVID-specific applications, like additional interfaces to smartwatches for pacing (Hansen, 2024). So, it is seen that the type of use or specific features of existing technologies are adapted to comply with the needs, wishes, and challenges of people with Long COVID.

Apart from active self-management during the day using technological tools, people with Long COVID also use existing technologies in another way; they use technology to improve their cognitive (executive) functions through brain training (Katella, 2024). An example of this is language training to treat brain fog using language apps like DuoLingo (Florian, 2022) or brain training games (NHS Health Research Authority, 2023; Research Team BEACON, 2023).

Critical conclusions

From the information on the current topic, we can conclude two points of critique on the current knowledge on the technology use of people with Long COVID to support the cognitive aspects of their everyday lives.

Current technology use in the rehabilitation of Long COVID and managing everyday life is usually not in the form of extending cognitive capacities by cognitive artifacts as described by Heersmink (2015). Mostly, technology in the form of devices, self-tracking, and learning interventions is focused on controlling physical symptoms and complaints or developing cognitive skills such as memory.

Besides, it is seen that current literature about technology use in supporting everyday life with Long COVID is often on the surface; the implementation in the everyday lives of people with Long COVID is

missing. An example of this is the focus on facilitation of regular care through digital solutions instead of integration and implementation of technology to directly support people in their daily activities (Daniels et al., 2024; Hawke et al., 2022; Rinn et al., 2023). Another observation is that literature on technology use in the context of Long COVID is often about the potential of wearable devices to track Long COVID-related symptoms through biometrics and the eventual recovery, while insight into the integration in everyday life is missing (Khondakar & Kaushik, 2023; Radin et al., 2024; Stewart et al., 2024).

2.3 Rationale study

Thus, Long COVID shuts down the lives of tens of thousands of individuals and causes a high societal burden. Despite the far-reaching impact, there is no solution for Long COVID yet. Therefore, the urgency is high to develop management methods for people with Long COVID so they can control their illness better and resume (part of) their lives. This thesis hopes to contribute to the visibility of Long COVID, gain insight into its impact on everyday lives by giving people with Long COVID a voice, and develop knowledge and ideas on these methods with a human-centered, qualitative approach.

The main research question is: *RQ: What is the (potential) role of technology in the management of cognitive aspects of everyday lives of people with Long COVID?*

This research question is divided into a few subquestions so that the relevant matters can be included in the eventual answer to the research question.

First, it is crucial to better understand what the impact of Long COVID on people's everyday lives looks like.

RQ1: How does Long COVID affect the everyday lives of people who experience fatigue, cognitive dysfunction, and eventual PEM-related symptoms that reduce their ability to carry out day-to-day activities severely?

As adjustments of cognitive strategies and the use of (technological) tools seem to be crucial in the management of everyday life, considering Long COVID symptoms, this is further investigated.

RQ2: How do people with Long COVID-related fatigue and cognitive dysfunction adjust and implement cognitive strategies and technology in order to manage the cognitive aspects of their everyday life?

To improve the management of Long COVID in everyday life and develop new interventions, it is relevant to look deeper into what is currently missing in the developed cognitive strategies and implemented technologies to be able to carry out daily cognition-involved activities.

RQ3: What do people with Long COVID currently miss in their developed strategies and implemented technological supporting tools to carry out daily cognition-involved activities and what can be improved?

2.3.1 Methodology literature

Before answering the research questions, it is important to get more insight into research about people with Long COVID (-related conditions), and suitable research methods for this user group. Based on these findings, the research method can be developed and eventually executed.

Research about people with Long COVID

Research into the impact of a condition or a certain intervention on everyday life needs the active involvement of the group of interest in the specific study. Yet, existing research about self-management interventions lacks recognition of people with Long COVID as the experts. These studies often focus on new interventions or (para)medical experts' opinions, but do not map the lived experience of people living with Long COVID who developed their own tactics and strategies (Dorronzoro-Zubiete et al., 2024; Rybkina et al., 2024; Stapleton et al., 2024; Weise et al., 2024).

However, people with Long COVID can be seen as experts in the field; they live with the condition and thus have a unique and valuable insight into how to cope with the complex character of their illness, and have already developed their own strategies and use of (cognitive) tools (Alwan, 2022). By ignoring the expertise of people with lived experience and only relying on traditional medical theories, research gets an ableist character (Alwan, 2022; Shew, 2020). This still happened in a lot of studies on Long COVID and other post-viral conditions, resulting in a lack of understanding of the condition and stigmatization (Alwan, 2022; Callard & Perego, 2021). Therefore, this study aims to listen to people with the lived experience of Long COVID, as to rely on their expertise.

Long COVID-adjusted measures for the method

Another crucial factor that should be considered in the study design is Long COVID itself; this condition (partially) disables people, resulting in restricted abilities to participate in studies. This results in two sets of measures for the study design, namely: 1) minimization of the impact of participation in this study for the participants, and 2) enabling participation despite participants' (potential) disabilities. It should be noted that these are closely connected.

1) Minimization of impact

This research will probably impact the participants because of their vulnerable health. Our group of interests is impacted by every type of activity, especially the ones that involve cognitive load. The effect can be, in the worst-case scenario, worsening of Long COVID complaints, like severe fatigue, headaches, and even decline of the disease (this is the case if people suffer from PEM, a common symptom of Long COVID). This depends on the severity of the disease, the activity pattern, and the specific complaints of the participant. Therefore, it is important to be aware of the vulnerable health situation of the participants during (the development of) the research method.

It is important that during the design of the research method, the researchers minimize the (cognitive) load as much as possible to reduce the impact of this study as much as possible. Besides, as people with Long COVID have varying disease profiles, personalization and optimization regarding symptoms need to be possible.

2) Enabling participation

Besides minimizing the impact, the design method itself should also be adjusted to the research group of interest to provide them with the opportunity to be able to participate in the study despite eventually limited capacities. The research method should complement this goal. A common method

to identify daily experiences and practices is an activity diary. The benefits of cultural probes like these are that they can be designed for a certain project and user group, the burden for researchers is relatively low, there is room for input, autonomy, and creativity from the participants, etc. (Carter & Mankoff, 2005; Gaver et al., 1999).

In a paper by Van Rijen (2025), several cultural probe methods in research with people who had a stroke were researched. The cognitive and physical impairments of people who have had a stroke and people who have Long COVID are partially similar. Therefore, this paper could be used as a guideline in designing the research methodology. To empower people to participate in research that is about them, the cultural probe should be adjusted to the needs of the group of interest to make it both useful and reciprocal (Van Rijen et al., 2025).

Approach to research method

This project aims to be participant-friendly. We are aware of the vulnerable health situation of the participants. The first researcher of this thesis also has Long COVID for several years and therefore has a good view of the risks and discomforts involved. Based on her experiences, deductive reasoning, and (in)formal literature, the research method is optimized as best as possible; in this way, people with Long COVID are facilitated to participate in this project and share their expertise with minimal impact on their well-being. We have tried to minimize the impact of this research as much as possible by adapting the method to the participants' needs, relying on participants' expertise, emphasizing autonomy and (social) safety, and adapting the method to the individual participants' pace.

The following section describes the research methodology that is used to answer the research questions.

3. Methodology

3.1 Design

This study is seen as explorative, and (mainly) qualitative. Using context information about the everyday lives of people with Long COVID, activity diaries, and semi-structured interviews, more insight into the potential role of technology in the management of cognitive aspects of everyday lives of people with Long COVID who experience fatigue-, cognitive dysfunction-, and (eventually) PEM-related complaints regarding their symptoms and eventual recovery was gained.

The starting point for the methodology was to research this topic in a human-centered manner; the expertise of the participants was leading, and we tried to minimize the impact of this project on the health of participants by adapting the method to the participants' needs, wishes, and challenges.

The study was approved by the HTI Ethical Review Board of the TU/e. We followed the guidelines for thematic analysis by Braun & Clarke (2013) for the design of the research method, the data collection, and reporting.

Contact with participants for the data collection took place from January 16, 2025, until March 11, 2025.

3.2 Participants

The participants were recruited through snowballing; via informal and formal (social) media such as LinkedIn and WhatsApp, using the (in)direct connections of the researcher(s), potential participants were reached. The recruitment material can be found in Appendix C.1.

This study focuses on people with everyday life-limiting Long COVID who experience fatigue, cognitive dysfunction, and eventual PEM-related complaints. Therefore, the participants had to experience these symptoms and indicate that the ability to carry out day-to-day activities is reduced significantly compared to before their COVID-19 infection. Another criterion was the duration of their condition; participants must have had Long COVID, either diagnosed by medical staff or by themselves, for at least half a year following the WHO definition of Long COVID⁸. This ensures that people have had the opportunity to develop management methods and strategies to cope with their condition regarding everyday life. Besides, participants needed to be aged between 18 and 67 (working age in the Netherlands) to ensure an active, involved life. Lastly, the study was conducted in Dutch, meaning that participants needed to be able to speak Dutch. Restricting to one language prevents mistakes, uncertainties, or ambiguities regarding (culturally determined) differences of interpretation of the questions and answers, and the transcription and data analysis are more coherent and transparent.

⁸ People could participate in case they have Long COVID symptoms that started within three months after (the recovery of) their SARS-CoV-2 infection for six months or longer. In practice, this means that the potential candidates ranges from people that were infected with the SARS-CoV-2 disease six months since the start of the data collection but never recovered from their initial symptoms resulting in Long COVID, or people that developed long-term symptoms three months after their infection in the first wave of the pandemic in 2020 and thus suffer from Long COVID for four and a half years during the participation in this study.

The initial goal was to recruit ten to fifteen people to participate in the study. During the data collection, the researchers determined the exact number considering the saturation of the data regarding the quality, depth, and variety. More than fifteen people contacted the researchers to participate, and eventually, twelve people started participating. One of them decided to drop out because of the cognitive load of the study design and health complaints. Thus, in total, eleven individuals were included in the study.

The participants were nine women and two men. The average age was 51 ranging from 27 to 63 years old. They experienced long-term complaints caused by a coronavirus infection 30 months ago, at the time of the data collection, ranging from 12 months to 50 months. Some participants suffered from other conditions, which were diagnosed before developing Long COVID, two got a diagnosis that was (possibly) caused by Long COVID, and others did not suffer from comorbidity. In Appendix B.2, you can find a more elaborate overview of the participants. Details in these descriptions were rephrased or left out, and names were replaced.

3.3 Setting

The first researcher conducted all the interviews and took charge of communication with the individual participants. She is a female human-technology interaction graduate student at the TU/e who gained experience in qualitative research during her studies. Although she had no active interview study experience, she did an interview training during her study at the university, and read the book on qualitative research of Braun & Clarke (2013). During this thesis, the first researcher had Long COVID for more than two (and a half years) manifesting in a range of symptoms, with fatigue, cognitive dysfunction, headaches, and PEM being the most prominent complaints. This means that she was part of this thesis's group of interest. This was taken into account and intended to be acted upon in a reflexive and transparent manner throughout the execution and development of this study. In Appendix B.1, you can find a positionality statement in which this reflexivity was more elaborately illustrated.

The other researchers were (i) an expert in human-centered, qualitative research within cognitive psychology and human-technology interaction, (ii) an expert in dementia-focused human-centered, qualitative research, and (iii) a design expert with experience in (activity) diary methods. They supported the first researcher with (the preparation of) the data collection and conducted code and theme checks during the data analysis.

Prior to the data collection, insights were gathered to optimize the research methodology. Firstly, feedback on the data collection was provided by one of the second researchers and an additional design and research expert. Then, a pilot study with four volunteers, of whom two had similar complaints regarding cognitive dysfunction and fatigue as the group of interest, was conducted.

3.4 Materials

Besides the materials of the three parts of the study, there was also context material, including the briefing and debriefing, and the informed consent form. These can be found in Appendix C.2, and Appendix C.3.

3.4.1 Sociodemographic questionnaire

The first part of the data collection was a sociodemographic questionnaire. This aims to collect data providing context about the participating individuals and thus describing the research context. The sociodemographic questions were both closed and open, ending with an opportunity to broaden the answers with an open field. These can be found (both in the Dutch and translated English versions) in Appendix C.4.

The form was made available online. Participants could execute this questionnaire in their own space and time and were asked to fill it in before the first researcher prepared the interview.

3.4.2 Activity diary

For the second part of this study, participants were asked to keep an activity diary for a week. This method aims to identify daily activities, the workarounds around these, the impact of Long COVID on the everyday lives of people, and the use of technology to support them with their condition. The activity diaries served two purposes: 1) They provided information on specific use of (the implementation of) cognitive strategies and technological tools which served as preparation data for the in-depth interviews, and 2) They were used as a probe during the interview provoking responses from the participants in the conversation about the concerned topic as described by (Gaver et al., 1999). The activity diary was developed aiming to meet the needs, wishes, and challenges of people with Long COVID to participate in research, as best as possible.

In case participants choose the physical option, they received an envelope by post at their indicated address, including a guideline, fourteen activity diary forms, and pens and markers to be able to fill in the activity diary. If participants preferred the digital format, they received the forms and guidelines via e-mail in PDF.

The guideline document contained a short explanation of the goal of this part of the study, instructions on the activity diary, and an example of every step of the activity diary. The activity diary included:

- 1) A text box in which participants could indicate the date;
- 2) The day diagram: A diagram in which people could sketch their fatigue, cognitive complaints, mood, and eventual additional health-related complaints over the day using (colored) bars and lines (time in hours on the x-axis, and smileys on the y-axis to indicate the severity/feeling);
- 3) Boxes around the diagram to describe the activities participants did, with three small boxes to indicate how well the activity went, whether the participant adjusted the activity to their Long COVID complaints, and to indicate the use of technology. Besides, there were lines to indicate the duration and to elaborate on relevant matters. They could connect the boxes with the moments in the diagram to indicate when they executed them.

- 4) The day evaluation: A box to evaluate the day using a 'grade of the day' ('cijfer van de dag' in Dutch) and questions to reflect on the day.

In the day diagram, participants were first asked to report on the course of fatigue, cognitive dysfunction, mood, and eventual other (health-related) matters during the day. Secondly, they were asked to report on at least two activities that involved (active) cognitive functioning regarding how well the activity went, the duration, whether they adjusted the activity to their Long COVID, the use of technology, and eventual additional comments. This reporting was done in a text box around the day diagram, which the participants connected with a line to the time it happened. Lastly, participants could add additional information in the diagram concerning, i.e., the use of technology, stimuli (like cognitive effort, light, and social interaction), rest moments during the day using drawing, words, stickers, or other (creative) ways to express themselves.

The day evaluation was used as an additional way to create insight into how participants dealt with the day (for both the participants themselves and the researchers). The questions were optional and allowed participants to share some extra thoughts and let them briefly reflect on the day.

The activity diary was executed starting approximately ten to twelve days before the interview. Participants were asked to report for seven days. The expectation was that the diary would take 10-25 minutes per day, depending on participants' characteristics and (Long COVID) complaints. As the expertise and autonomy of people with Long COVID were leading during this study, the researchers did not oblige participants to fill in the form at a certain time in the day but instead let them free. The activity diary needed to be finished a few days before the interview moment so the first researcher was able to process the information in preparation for the interviews as described by Carter & Mankoff (2005). The materials for this second phase of the data collection can be found in Appendix C.5.

3.4.3 Interview

The third part of this study was interviewing. This step aims to complete the data that is needed to answer the research question. The interviews elaborated on the activity diaries, and the sociodemographic data were taken into account for context. The principle was to have an open conversation (so not a strict interrogation) about everyday life with Long COVID, focused on the topics of the research questions, dividing the control between the researcher and the participant.

The interview schedule was defined by the first researcher and, with the guidance of the second, more experienced researchers, further developed to ensure quality. The schedule contains a briefing, a semi-structured questionnaire, and a debriefing. The list of questions is semi-structured, which means that potential additional questions and suggestions based on, e.g., earlier contact and the activity diary forms were prepared and added to the interview scheme besides the main questions. Also, these questions were not necessarily asked depending on the input of the concerned participant. Table 1 shows the (translated to English) interview questions⁹. The original Dutch interview schedule can be found in Appendix C.6.

⁹ Note that in some of the interviews, question 7 was changed into "What do you currently miss to be able to perform everyday tasks in combination with your Long COVID?" depending on the answers on previous questions.

Interview questions

1	How are you?
2	How did filling in the activity diary go?
3	Could you please explain to me how you filled in the activity diary?
4	What did you notice?
5	How do you approach an activity that requires thinking when you notice that your brain has difficulties functioning? (for example, because of fatigue or overstimulation)
6	How do you currently use technology?
7	What do you miss in the technology you use or which you know exists to enable perform everyday tasks that require thinking in combination with your Long COVID?

Table 1: Interview questions

The online interviews were conducted via video calls in Microsoft Teams, and the live interviews were recorded with Microsoft Teams. This software was used in both online and live interview settings to record and transcribe the interview. As TU/e-students and employees have a secure account, Teams can be seen as a safe platform for research activities.

Nine interviews took place online, and two interviews were conducted at participants' homes. The interviews could include breaks depending on the individual needs of the participants, and they could eventually be broken into pieces to overcome the barrier of Long COVID-related restrictions to participate in research, as described by Heaton-Shrestha et al. (2022). Another measure that was taken to optimize the involvement of participants was the active proposal to add information via email after the interview if participants felt they had more insights after some time.

The interviews took 36-60 minutes, with an average duration of 50 minutes. All the interviews were conducted in one take as participants preferred this, and none of them added information after the interview.

3.5 Procedure

After people contacted the researchers about their potential participation, they received a message with dates to plan a call with the researcher for the briefing. During this call, the researcher briefed the potential participants on the content and course of the study. This conversation also included discussing preferences (interview date, and choice of activity diary) and the needs (additional adjustments to the study method) of the participant to be able to participate in this study. After the call, people received an email with the briefing repeated, and they were asked to read the consent form carefully (this can be found in Appendix C.3), sign it, and send it back to the researchers.

After participants gave their consent, had no further questions, and were aware of their rights, the study could begin. The participants received an email message with a link to the sociodemographic questionnaire and were invited to answer these questions at least two days before the interview took place. After opening the link, the questionnaire in Microsoft Forms appeared, and participants answered the open and closed questions. Lastly, they were asked whether they wanted to share anything that was not discussed in the questions.

In case people preferred the physical option, they received an envelope with the activity diary instructions, fourteen forms, and the accompanying materials. A few days before the activity diary phase started, participants received a reminder via e-mail with a note that they could ask for help

with the activity diary if needed. In the case of the digital option, the activity diary instructions and form were attached to the message.

The next step for participants was to get started with the activity diary. Starting ten to twelve days before the interview, participants kept this diary for a week. During this week, the researchers sent a message to the participants to remind them and to offer support. For each of the days in this period, participants were asked to report on fatigue, cognitive dysfunction, mood, eventually additional (health-related) complaints and reflect on at least two everyday activities that involved cognitive functioning during the day, and eventually evaluate the whole day by a day evaluation. After participants finished the activity diary for the determined period, they shared their activity diaries with the researcher by emailing them. The physically filled-in diaries were scanned or photographed. The researchers sent an acknowledgment of receipt and asked for possible extra pictures in case of unclarity. Besides, participants were reminded of their interview date and time, and the settled agreements on breaks and adjustments to comfort the participant.

At the start of the interview, the interviewer checked in on the participant, and a short briefing was given by the interviewer. If the participant had no further questions and agreed with the start of the recording, the researcher started the video and audio recording on Microsoft Teams. The interviewer followed the interview schedule with additional or adjusted questions, based on the activity diary.

When all the main questions were answered, the interviewer asked whether the participants wanted to add anything to the information they had already shared. The interviewer informed the participant about the option to share any additions via email when their Long COVID has restricted them too much. The debriefing was given, the interview was closed, and the recording was stopped. Participants were told that they could share any additional information afterward via email, phone, or WhatsApp. The researcher also offered to share the findings at the end of the project (see Appendix F) and to eventually share the published thesis. In Figure 2, the timeline of an individual participant's procedure (including contact moments) is shown.

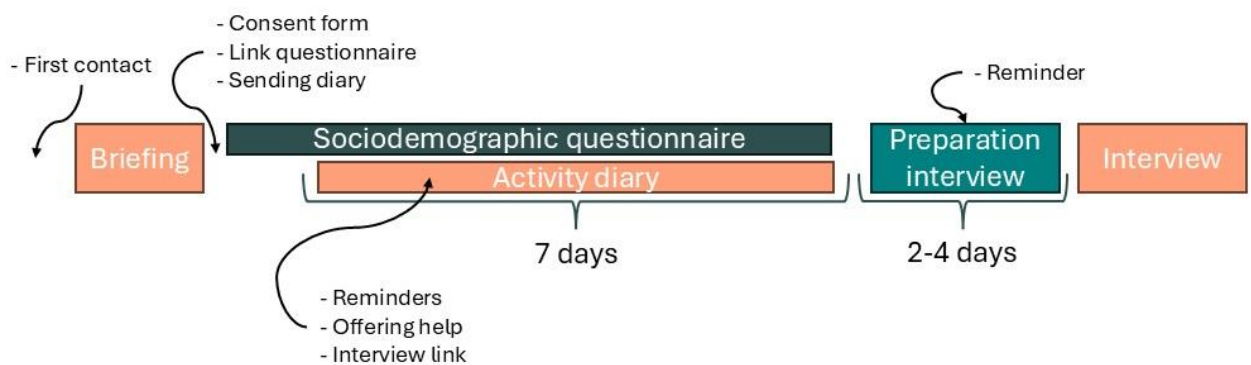


Figure 2: Timeline of the procedure

3.6 Data analysis

The data analysis consisted of three main components: defining the context of the participants, describing the observations of the submitted activity diaries, and a thematic analysis to answer the research questions.

3.6.1 Context participants

The context of individual participants was described using the data from the sociodemographic questionnaire, the activity diaries, and the interviews, serving as a definition of the research sample and as background information for (future) researchers. The aim was to provide insight into the participants' Long COVID stories and background but also protect their privacy. Therefore, only relevant information was included, potentially recognizable data was described vaguely, and the names were replaced with other names. The descriptions can be found in Appendix B.2.

3.6.2 Activity diaries

Together with the diaries themselves, the notes of the diaries that were made by the first researcher as preparation for the interviews were used to describe how participants handled this part of their participation. By means of the content, the amount, and depth of the activity diary data, the range and differences between the diaries were described. Besides, during the interviews, participants were asked about their experiences filling in the activity diary. These were evaluated using the thematic analysis approach and were discussed.

3.6.3 Thematic analysis

A thematic analysis was conducted to analyze the lived experience of people with Long COVID, providing in-depth information on the workarounds and use of technological tools in everyday activities that involve cognitive functioning, using the interviews and descriptions of participants. The instructions of Braun & Clarke (2006, 2013) on thematic analyses were followed. The analysis was performed using the qualitative software program NVivo (Lumivero, 2024).

After the execution of the interviews, these were individually transcribed by the Teams transcription function and complemented and adjusted by the first researcher, comparing the provided text and audio recording closely. In this way, the data was transcribed efficiently and thoroughly, while the first researcher also familiarized herself with the data.

During the coding process, the first researcher identified anything of interest and relevance. This took place during and after the data collection. The coding was discussed with one of the second researchers, comparing two coded transcripts. As a third step, the first researcher sorted the codes on a whiteboard, developing patterns, and later on defined the themes (the final version can be found in Appendix E.1). As this research can be seen as explorative, the themes were based on the data itself (inductive approach). To increase the quality of data analysis regarding validity and reliability, the themes were discussed with the second researchers who were (partially) familiar with the data. This also contributed to a broader and diverse perspective on the data. Based on this input, the themes were adjusted, complemented, and eventually optimized.

As the last step of the data analysis, the final themes were defined and presented in the results section. In combination with the context information of the participants, these were discussed critically and in-depth using quotes from the participants (translated to English).

4. Results

4.1 Observations from the activity diary

The activity diaries of the participants differed substantially, which complies with the free instructions and differences between personality types. In this section, the similarities and differences are discussed. Participants' experiences are shared in the next subsection on the thematic analysis.

The activity diary data differed **content**-wise; some participants focused especially on cognitive activities while others focused more on everyday tasks that impact the course of the rest of the day, including physical and cognition-involved activities. It was also seen that some people mentioned their rest moments, the impact of an event on their mood, and the complaints they experienced during or after the activity.

Some substantial differences were observed regarding the **amount** of information. The number of activities participants documented differed between one to two activities per day, and information on four to five activities. Also, the level of detail in the diagram differed; data in some of the diagrams were tracked per hour, while in others, more loosely documented. It was also seen that some participants included additional features in the diary, like health-related issues in the day diagram, and others indicated their rest moments, use of technology, or other events. This indicates that participants made the diary their own.

Besides, the **depth** of the information also differed between the participants. Some participants elaborated on the activities by providing information on their experiences, feelings, adjustments, or context, or explaining certain choices. Something similar was observed in the day evaluation.

Another conclusion that can be made from the activity diaries is that the course of complaints, mood, and other condition indicators differs. For some participants, fatigue and cognitive dysfunction go hand in hand, while others experience a disconnection between these two. Also, for many participants, mood and health-related complaints seem to be separated from each other. Besides, in the set of activity diaries, it can be seen that the input of individual participants differed between the days they filled in the diary.

Lastly, it should be noted that one of the participants only documented five days instead of seven days divided over a longer time because of the confronting effect on the participant. Another participant used her own method as she did not notice that there was a prepared form and guidelines to follow.

4.2 Thematic analysis

The data collection for this thesis was done in Dutch. Therefore, the quotes were translated into English, referring to the original ones, which are presented in Appendix E.2 with numbers between brackets ([x]) at the end of every quote. The quotes are translated and paraphrased considering meaning, form, and readability for the sake of the accessibility of the report and authenticity of the quotes. Details in the quotes that contribute to the recognizability of the participants were rephrased or left out of the original and translated quotes. The names of participants were replaced by other (random) names, and an overview of them was given in Appendix B.2: Descriptions of the participants.

The thematic analysis resulted in nine themes, of which seven are also subdivided into subthemes. These themes can be divided into three parts: a) the current situation, 2) coping with the situation, and 3) the use of (technological) tools. The themes and subthemes are presented based on their category in Figure 3 and further elaborated on in this section.

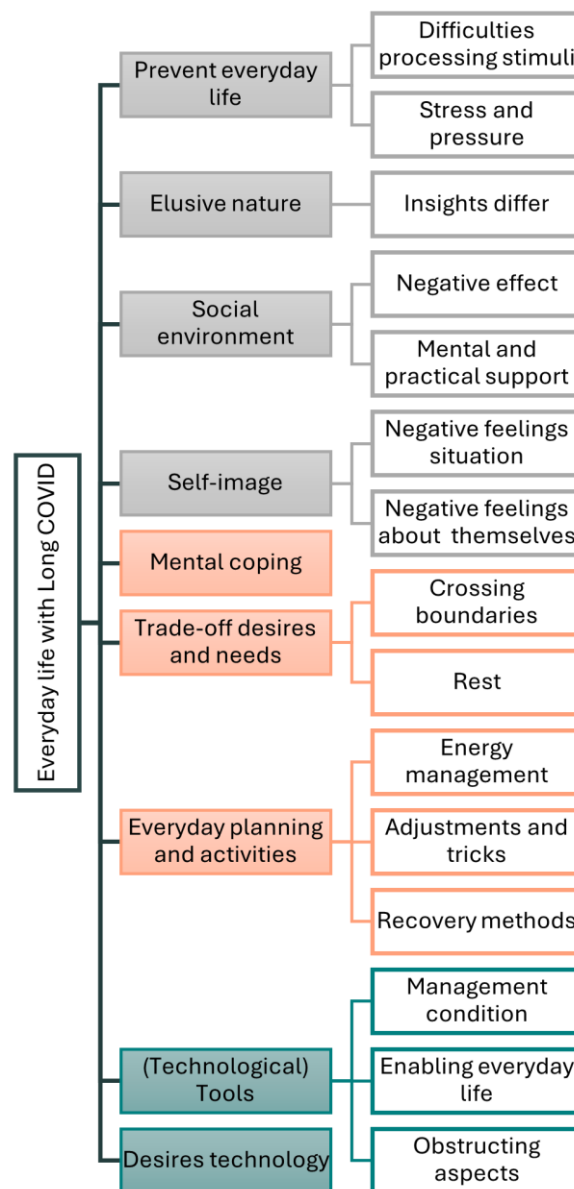


Figure 3: Themes and sub-themes

4.2.1 Experiences with the activity diary

Before introducing the actual themes of this thesis, participants' experiences with the activity diary are discussed in this subsection.

The participants experienced the activity diary part differently. For some of them, it was cognitively demanding to fill in. From this, it can be concluded that despite the Long COVID-specific adjustments to the design of the activity diary form, the cognitive load was still experienced as substantial. Besides, several participants shared that they found it difficult to fill in the form; apart from cognitive load, people seem to have experienced a certain mental load in filling it out correctly. A few participants explicitly stated that they experienced both cognitive and mental load while filling in their diaries.

Apart from this, people were also positively affected; almost all the participants indicated that the activity diary provided new insights into their Long COVID regarding everyday life. Especially the impact of activities on participants' health-related complaints became clearer because of the diary, which led to new insights on how to cope with their condition. Bente explained: *"So that's what makes it interesting to consciously reflect on things that take energy. And that also let me think like how can I prevent that I drain so much on these things?"* [1]. Some of them also concluded through filling in the diary that they experience that they are doing too many things regarding their condition.

Other participants experienced the diary as confronting because they realized that they are more impacted by Long COVID than they were aware of before filling it in. An example of this confrontation is Daphne, who has had Long COVID for more than four years but is (partially) actively involved in society. She stated, *"I found this more confronting than expected. And indeed, you make more adjustments because you do these automatically already, which also felt confronting because I often have the feeling that things are actually going great, and I can do a lot of things, I am doing a lot of things, and I am present with many things. But that means that I go to bed at 8.30 pm or I am not able to sleep because my head is all over the place. That is confronting for me."* [2]. This also emphasizes the relevance of the activity diary in the research methodology and reflects on the reciprocity of the diary.

It can be concluded that filling in the activity diary was a substantial and intensive process that led to more insight into one's condition and potential strategies for most of the participants. However, some people also experienced mental and cognitive impact, although it should be noted that that is not per se something negative.

The following themes present how people with Long COVID experience the impact of their condition on everyday life. This includes how their current situation looks like regarding activities, the manifestation complaints, their social environment, and their mental self-image.

4.2.2 Theme 1 – Prevent everyday life

The first theme identifies the experience of all participants that **Long COVID prevents people from participating in (cognitive) everyday life**. Because of symptoms and health-related complaints, (cognitive) functioning is impaired. This leads to difficulties with executing everyday tasks and thus disable involvement and participation in everyday life. Noortje summarized this impact the following: *“Well, you cannot live the life you want to. I cannot go to work, I cannot go to birthdays, I cannot do any crowded things.”* [3].

In certain situations, people’s health-related impairments not only lead to difficulties with everyday activities but even result in (cognitive) crashes and errors. These can be small errors that can also be solved, but they can also be more substantial errors that lead to crashes of (cognitive) functioning.

Cognitive dysfunction can manifest, among others, in difficulties with basic cognitive functioning. More than half of the participants explicitly indicated that they have difficulties with certain essential cognitive functions that impair their ability to execute everyday tasks. This includes functions like reading, concentration within an activity, and (forward-)planning. Besides, almost half of the participants experienced that certain cognitive functions are explicitly impaired because of their Long COVID. They verified this through measurable/quantifiable activities, such as reasoning capacities through puzzles or language and writing through e-mails.

Basic cognitive functions are essential for functioning well. This shows that people with Long COVID can experience difficulties with cognitive functioning, which hinder active participation in (societal) everyday life. This also aligns with the following sub-theme:

Sub-theme 1.1 – Difficulties processing stimuli

Every participant indicated that they **find processing stimuli during an everyday activity challenging and can disable one**. As information in any form is involved in all kinds of activities, it is crucial to be able to (actively) handle stimuli. Otherwise, you are shut down from everyday life.

This pattern is related to several types of stimuli processing; the information within the activity that needs to be processed to execute the activity properly, and stimuli in the environment that complicate the process of the activity. First, the experiences regarding (relevant) information within everyday tasks are discussed.

Eight participants indicated experiencing difficulties with information processing relevant to the activity. This means that they are not able to understand the information, that they are not able to convert this into the right action, or both. This can lead to, for example, difficulties with reading the newspaper or processing traffic while driving.

Especially activities with many steps seem to lead to complications; five participants indicated that they have trouble doing multiple things within or during one specific task, like household chores and personal care. Veerle explained that this arises for her during her showering process: *“It is just really weird, because you would think: what is difficult about showering? But apparently, I find I very hard. There are a lot of steps involved; you have to go to the bathroom, you have to take off your clothes in*

a certain order, you have to go under the shower, the water should have a certain temperature, then you have to think about washing your hair, shampooing, dry off, put your clothes on again.” [4].

Consequently, this can also lead to difficulties in multi-tasking, which includes multiple separate cognition-involved processing efforts within one activity. In addition, it should be noted that because of difficulties with processing, filtering relevant information can be tough as well.

Besides, several people experience that disruption of their activity process can be disadvantageous for the activity execution or even harmful for their course of complaints during the day. This disruption can be caused by small interruptions within an activity. You could argue that these disruptions within activities can be seen as an additional effort to another activity, which can be associated with multitasking. Daphne experienced that an everyday activity can be complicated through (external) interruptions: *“I have more difficulties with expressing myself in e-mails; when I have a phone call in the meantime, and I came up with a good sentence in my mind before, it is gone, I lost it.” [5].* Besides, deviations in routines are related; deviation from one’s regular process increases the cognitive load, which, apparently, is experienced as stressful or leads to errors.

Apart from the information from within the task, people also experience that their functioning is affected by external stimuli, including, e.g., other people, sounds, and decision processes. Daphne described this for a certain event: *“I still find it very unpleasant if I get overwhelmed by crowds. And if I should enter the room between all these people, and also think where I want to sit, that won’t work well. Then I’ll get an error, that just demands too much of me.” [6].* These external stimuli can restrict people from doing certain activities or going to certain events.

Sub-theme 1.2 – Stress and pressure

Many participants indicated that they experienced **having difficulties handling (mental) stress regarding their condition**. They experience that it costs them more to cope with stress. Roos explained: *“If you are in the car and you still have to drive for half an hour, but you have to pee, you experience pressure, stress within your body. So, it’s in the things that lead to unrest in your body which get you tired more quickly.” [7].* Some participants experience a substantial decrease in energy after they have had stress. It should be noted that even something small can cause stress; it seems that people with Long COVID are triggered and handle stress differently.

As stress and any other kind of pressure are part of everyday life, this finding emphasizes the challenges people with Long COVID can experience regarding everyday life.

4.2.3 Theme 2 – Elusive nature

This theme demonstrates **that participants experienced that health-related factors are interconnected, which contributes to the elusive nature of Long COVID**. If one experiences fatigue or another complaint, this triggers new Long COVID-related complaints (which are not necessarily directly related). Roos explained: *“But if I get cognitively overstimulated, I also notice this in my body.” [8].*

This is especially observed regarding energy. After exertion, many participants experienced complaints. Mismanagement of exertion can be manifested in complaints for a few hours, but also for multiple days afterwards, depending on the level of impact and the individual’s condition profile. Noortje explained how she experienced this pattern: *“That Monday, I pushed my limits hard. Especially the conversation about my little son, which was really necessary, but the day after I was*

not feeling well at all. And that Wednesday, it was even worse. I canceled meetings and I had to lay down in bed all day.” [9].

Besides, energy level, other medical conditions, and sleep quality are associated with complaints and tolerance regarding stimuli. As Veerle explained: *“If I have limited energy, it goes less well cognitively. I notice this while expressing myself verbally, having a conversation etc. For example, I am especially bothered by sound.” [10].* A significant part of the participants indicated that when they have difficulties with processing stimuli because of fatigue or overstimulation, they can crash.

Sub-theme 2.1 Insights differ

This sub-theme describes that **the insights into complaints differ both between and within people.** It seems that people with Long COVID have developed an understanding of how their bodies react. For instance, indicated that Henk did not learn anything from the activity diary, as he already has developed an understanding of this illness, as he explained: *“I haven’t learned anything. For me it is a recognizable pattern that keeps coming back. It is going well, so a few things to make it more positive and you have one action that costs a lot of your strength and afterwards everything declines.” [11].* These ideas are mostly not based on any scientific knowledge, but more on their experiences and observations.

However, some participants experience the manifestation of their Long COVID complaints as unpredictable. Noortje pointed out this mismatch of one’s perception and the manifestation of complaints the following: *“Sometimes things feel differently as they are in the end. So, you think you have more energy than you appear to have.” [12].*

Some experience both an understanding of the course of their condition but cannot always rely on this insight because of the unpredictability of their complaints. This contributes to the elusive nature of Long COVID in everyday life.

4.2.4 Theme 3 – Social environment

This theme describes **that people with Long COVID have different experiences with their social environment.** They appear to have distinct feelings towards the involvement of others in their Long COVID process and their everyday life, differing between and within individuals. On the one hand, people feel supported by other people, while on the other hand, it can lead to negative feelings. Therefore, this theme is divided into two sub-themes describing the negative and positive implications of social contact.

Sub-theme 3.1 – Negative effect

Ten of the eleven participants experienced that **their social environment can have a negative effect on their energy or mental status.** This can be divided into two categories. On the one hand, six people experience misunderstanding regarding Long COVID and their specific situation within their social environment. People experience this as a disconnect between themselves and their social environment. Some of them explicitly indicated that they are negatively affected by this. For example, Daphne explained: *“Nobody realizes it. I can still be affected if people ask, ‘Does Covid still exist?’ I find that very difficult. And also, people that say, ‘I just go to work, I have been sitting at home long enough now’, I find that hard.” [13].* Another negative aspect of the social environment can be the impact on one’s energy; more than half of the participants experienced that people

around them can be seen as energy consumers. Veerle experiences this, she pointed out: *“After people have been gathered together, I am totally drained.”* [14].

Sub-theme 3.2 – Mental and practical support

The other side of social contact is that **people’s social environment serves as mental and practical support regarding their Long COVID**. Especially, the mental impact can be substantial; receiving understanding from people around them is experienced as supportive for their well-being and condition by many participants. Sharing one’s experiences with their condition with others can be mentally helpful, but participants have also found peace of mind in the compassion of others.

Another side of the involvement of one’s social environment can be practical support. Because of supportive coaching and thinking with them, more than half of the participants experience that this practical support of their peers enables certain everyday life activities. While Maaïke executes activities like cooking with her husband, as she remarked: *“So we often do it together. We have a nice cooking island so often I am on one side, and he is on the other side. My husband is cooking, keeping an eye on it and I am cutting. So that’s how we do it these days.”* [15]. So, practical support can manifest in several ways and can enable one to participate in everyday life.

Also, by taking over tasks or backing up, some participants experienced social support as beneficial. The presence of another individual can provide a certain type of confidence (by acting as an opt-out version), enabling the person with Long COVID to do a certain activity.

Lastly, more than half of the participants experience that social contact is good for one’s mental condition; people get energy from being involved in social contact, which is also beneficial for their overall well-being. Anouk remarked that her mental battery charges through social contact: *“If I have a bad day, I am really bummed out. But I get a lot of energy from people visiting me or my son, or good weather.”* [16].

4.2.5 Theme 4 – Self-image

Because of Long COVID, people experience a change in abilities and energy. As a result, one’s identity has changed, work has reduced, and responsibilities have decreased. A large majority of the participants experienced a change in their self-image, directly or indirectly. **So, besides the physical effects, Long COVID also impacts the mental aspects of one’s self-image.**

Noortje described the effect of her health-related complaints on her identity the following: *“Your willpower, your passions, everything is gone.”* [17]. Veerle also discussed a similar experience: *“I am not sure what happens. Why you have done your master’s, have a really analytical job, but then cannot think that you first need to take off your slippers before you put on your trousers.”* [18].

Besides the change regarding their self-image, participants often looked negatively at their current situation regarding their activities and abilities. This can be divided into two sub-themes describing the negative feelings towards their current situation and themselves.

Sub-theme 4.1 – Negative feelings situation

This sub-theme describes **that some participants think negatively about their current situation**. They experience disappointment and a sense of nostalgia or melancholy towards their life and abilities.

On the one hand, getting Long COVID has felt unfair to some participants. Some participants stated explicitly that they were upset by their Long COVID, especially during the first period. While other participants seemed to be more actively disappointed about the things they cannot do anymore because of their condition. This negative feeling can also be expressed through melancholic feelings towards their (healthy) past as Karin explained: *"I did everything on my own at home, odd jobs around the house, the garden, everything. How could I have ever done it?"* [19].

Sub-theme 4.2 – Negative feelings about themselves

Although some people had negative experiences with their current situation, the majority of the participants developed negative feelings towards their current abilities and activities. This sub-theme describes the fact that **most participants think negatively about themselves and seem to condemn themselves**. This especially comes forward in the way people look at their current identity; they feel and/or think that their activities and presence are less worth it. These feelings can be manifested within activities, as Bente explained about her work: *"Because I want it all to succeed, I do not give in to take a ten-minute break doing something else. I just want to do, like the rest, don't take any breaks and continue."* [20]. But these can also be experienced regarding the things one does; some participants seem to look down on the activities they do that comply with their needs and desires.

Another comment some participants make is that they choose to do 'lazy'. Anouk explained: *"You look for the easy things, you become complacent. Almost lazy for some things, but because you have to for many things."* [21]. This implies that despite being aware that they need to do less because of their condition, they condemn their choices or abilities.

This is also observed among some participants who have the feeling that they should be able to do certain things and experience stress to do good. You could say that the expectations people and others have towards them do not always match the reality of Long COVID.

Thus, it seems that a lot of the participants (indirectly) condemn their current selves; although they are aware that they are not responsible for their situation and do not have meet certain requirements, they still experience a negative feeling towards the things they are able to do, and who they currently are (because of their condition). There seems to be a mismatch regarding their ratio and their feelings.

It is important to note that the impact of Long COVID on participants' mental state also manifests in stress responses or complaints. Some participants experience this when they talk about the topic, for example, during their interview for this study. Daphne pointed out that she experiences complaints when discussing her condition: *"I am noticing, I'm already feeling it now, when talking about it, especially COVID, somehow my breath goes the wrong way. I am not sure why, but it feels like it is too much indeed."* [22].

Thus, Long COVID not only affects the cognitive and physical aspects, but also one's mental state can be impacted. Although most participants seem to have a (medium) positive mood, they suffer mentally from their condition to some extent.

The upcoming themes describe how people with Long COVID cope with their current situation. They have developed methods to live their everyday life as best as possible through mental coping mechanisms, trading off desires and needs, and (cognitive) strategies.

4.2.6 Theme 5 – Mental coping

Many participants experience that it is crucial to invest in methods to cope with their condition and the associated mental impact. Therefore, **people actively implement mental coping mechanisms to enhance their mental well-being.**

A major group of people indicated that accepting one's situation was crucial for recovery and coping with their condition in everyday life. As Veerle explained: *"I just think 'What a nice weekend, I'm feeling very happy. And I feel awful physically, but I am still able to lie down in my bed. I have a nice bed, there are only a few things I really have to do so I can take that rest.' I have come to accept that that's the way it is."* [23]. This resignation seems to be fundamental for the mental state of many of the participants and their overall quality of life.

Another mental coping mechanism was to enjoy the (little) things in everyday life. This does not necessarily come naturally; several participants indicated they actively work on this. Lotte remarked: *"Well, I am actively working on it to not see the glass half-empty but really half-full. So, I am working hard to enjoy the little things, the things I do have."* [24].

Some participants also found strength through comparing their health-related situation with stories of people who have more severe Long COVID. It seems that imagining worse scenarios downgrades the impact of one's own condition. As a result, they are better able to handle their situation mentally.

To mentally cope with Long COVID, some participants also receive mental support through therapy. This can be psychological treatment, but also mind-body therapy was mentioned by a participant, and ReAttach therapy.

So, although many participants experience a positive mood, they have to actively implement methods to cope with the mental impact of Long COVID. It seems to be hard work to stay or become happy living with such a condition.

4.2.7 Theme 6 - Trade-off desires and needs

This theme describes that **Long COVID forces people to constantly weigh their desires against their (health-related) boundaries and rest.** This balancing act is a constantly present issue in their everyday life.

Sub-theme 6.1 - Crossing boundaries

Despite their insight into their complaints and the course of their condition, almost all participants experience that it is sometimes necessary to go over their limits to be involved; **living and enjoying everyday life force people with Long COVID to cross over their (health-related) boundaries.** Henk also indicated this during the interview: *"As I said, I am a bit fatigued because during the weekend I went to a birthday party for a few hours, and I feel that for two to three days afterwards. But I have that experience, so I knew this was going to happen. But that doesn't mean that I didn't go to that birthday; I have to participate in social life, otherwise nothing will come of it anymore."* [25].

Crossing one's boundaries is often related to efficiency or enjoying activities; people with Long COVID experience that it is sometimes needed to go over their health-related boundaries to do nice things and keep up with the pace of everyday life.

Being involved in experiences outside one's Long COVID bubble often contributes to mental well-being, as five participants explicitly mentioned. Roos remarked: *"You make the exceptions for things that make you happy, that produce happy chemicals, and this is one of those."* [26].

Another pattern that is observed in this theme is that crossing boundaries often takes place in responsibility-related situations; responsibilities that people with Long COVID have do not comply with the needs of their condition. Noortje explained: *"That really wrecked me for about two years. It has not been good for my recovery, because things have to happen. Even if I feel really bad, I am still bringing the children away and if I feel even worse, I am still cooking and if it is really too much, I am building marble runs. Because you have to, my husband works, so a lot comes down to me."* [27].

Besides, some participants indicated that they sometimes choose strategies to handle a certain situation out of comfort, while this is not per se beneficial for their condition.

Thus, participants are often aware of what is good for their condition. But to participate in everyday life and for their mental well-being, they cross their boundaries; sometimes they choose to do so but, in some cases, situational characteristics determine this for them.

Sub-theme 6.2 – Rest

This subtheme describes that all the participants are thinking about implementing rest in their everyday lives to some extent. **So, the concept of rest is always present in everyday life with Long COVID.** Most people indicated that they take rest during the day to survive the day and to be able to do activities.

However, the impact of rest and what rest looks like differed between people and also within individuals. Some participants do not recover from taking rest as Veerle pointed out: *"Besides, I also just discussed this with my husband about resting and restoring. I have to rest, because I cannot stay up all day. But when I lie down for half an hour after I have been staying up for some time, I feel energized again."* [28]. While for others, a rest moment can restore their energy level.

Others don't want to take rests or have difficulties with taking enough rest moments. Henk remarked that he does not want to take rest the usual way: *"I like to go into nature and walk, because, as I always say, my battery can charge again. I don't lie down on the couch to sleep, which doesn't work, and I don't want that."* [29]. And others, like Wout, are focused on the quality of sleep.

This indicates that everyone is preoccupied with the concept of rest related to their condition; it seems that people experience some kind of pressure regarding resting because of their condition. Therefore, this sub-theme can be interpreted as advice for future research to look into how people experience (the concept of) rest regarding their condition.

As a result, people with Long COVID cross their boundaries to live their everyday life, but on the other hand, need to consider their health-related complaints, and they need to have some kind of rest during the day. This act of finding balance between (mental) desires and health-related issues can be related to theme 3 about the trade-off between the negative and positive aspects of social contact.

4.2.8 Theme 7 – Everyday planning and activities

This theme describes **how people with Long COVID adjust their everyday lives regarding their Long COVID**. This includes (practical) adjustments within everyday activities regarding time, stimuli, (cognitive) effort, and capabilities, but also adjustments that contribute to executing cognitive actions within activities.

Sub-theme 7.1 – Energy management

This sub-theme shows that **participants indicated that they adjust their everyday planning regarding their Long COVID complaints and energy**. Some participants distribute activities throughout the day. But also, more than half of the participants consider (expected) complaints. For example, Veerle, who adjusted her planning on Mondays because of exertion in the weekends with her children, as she told in the interview: *“Also, after the weekend on the Mondays, I never plan anything because I have to take it easy.”* [30]. And some participants explicitly stated that they plan only a limited number of activities during the day. Also, structuring everyday life seems to be a way to handle their (health-related) needs.

Sub-theme 7.2 – Adjustments and tricks

All the participants recognized that they are not able to execute certain activities without additional strategies; their (cognitive) abilities do not match the original task. **Therefore, they (actively) implement adjustments or tricks to enable themselves to execute (everyday) activities**. These methods are related to minimizing cognitive effort, decreasing the impact of stress, and setting boundaries. The most indicated solution is to keep activities simple and easy. But people also implement other tricks so activities, like cooking or meeting with people, comply better with their needs.

A specific strategy is the use of lists or notes. A significant majority of the participants use lists or notes to support their thought process. This method is mostly used to extend one’s memory; people experience that lists or notes can serve as a reminder tool for everyday details and events like things to do during the day, the content of a conversation with a friend, or groceries. Lotte explained this the following: *“Writing a lot of things down in my diary, also reminders for myself. And often I also set an alarm. But especially things people tell you, for example, a friend who goes on a ski holiday in Austria next week. I write that down, otherwise I just lose it.”* [31]. Two participants use lists to structure their thoughts during an activity or to prepare for a certain event. Bente indicated: *“I try to prepare these types of meetings, and I notice that I am a lot in my mind. So nowadays I write everything down in Notion, what I come up with, to structure everything a bit.”* [32].

Sub-theme 7.3 – Recovery methods

To recover from Long COVID or event-related complaints (and relapses), **many participants indicated that they actively implement strategies and methods for recovery**. This theme has a broad range. Some people train their cognitive functions, which contributes to their return in everyday life and related activities.

Others did more substantial implementations in their life, like following treatment methods or using medication, which contributes to recovery. Lastly, nature and walking are experienced by many participants as healing, contributing to recovery. Wout illustrated this: *“So I draw a lot of energy from that during the day and deliberately take rest during the day, walking at a very slow pace, slowly*

taking your steps. Not because I cannot handle it physically, but because that rest in pace also provides rest in my head.” [33].

Thus, it can be concluded that people with Long COVID actively anticipate their (health-related) complaints and abilities through implementing strategies. These enable them to execute everyday tasks. Besides, they are keen to improve their situation through developing and looking for recovery methods and strategies.

The last two themes of the thematic analysis describe the (potential) use of (technological) tools. This includes the current use to manage or enable everyday life, but also the experiences of desire for technology to support people with Long COVID.

4.2.9 Theme 8 – (Technological) tools

This theme describes that **(technological) tools play a role in the management of Long COVID and enable and involve people in everyday life. However, there are also complaints regarding (the use of) technology** in this context. The current role of technology and tools is divided into two sub-themes, and the objections and obstructions regarding technology are presented in a separate sub-theme.

Sub-theme 8.1 – Management condition

Almost all the participants use (technological) tools in the management of their condition to some extent. Technology is especially used by people to get insight into their condition. An example of this is Noortje’s smartwatch that brings her attention to her physical efforts as she pointed out: *“But also without realizing, because you don’t feel it that way at that moment, but at a certain point it says: ‘you’re working out well’ And then I’ll think ‘how? I wasn’t, but thanks’.” [34].* In some of these cases, people experience it to be valuable to adjust their planning or activities in response to the technological output to manage their complaints.

Also, (non-technological) tools to prevent overstimulation are often used to prevent overstimulation. And in some cases, people have experienced (technological) tools beneficial for the recovery of their condition. This can also be controlling comorbidity through technology, contributing to the improvement of one’s overall condition.

Sub-theme 8.2 – Enabling everyday life

This sub-theme demonstrates the role technology plays in the facilitation of everyday (social) life for people with Long COVID; **Technology enables and involves people in (societal) everyday life (activities)**

For some people, technology is fundamental in the connection with the external world outside one’s Long COVID bubble. On the one hand, technological interventions can serve as a provider or support for communication between a person with Long COVID and someone else. Besides, technology can also serve as mental support. Noortje experiences her Google Home as a provider of social support during her days alone at home, as she explained: *“I like it. I also like that it speaks, then you’re feeling less lonely [laughing awkwardly]. It is kind of nice.” [35].*

Another aspect of (technological) tools is enabling one to execute activities, through supporting the physical or cognitive aspects. A majority of the participants use technology to support them with cognitive tasks. This can be related to supporting information processing, but also other cognition-

involved tasks like cooking. Wout uses adaptive cruise control to support the cognitive aspects of driving as he: *“So I don’t have to put energy in that to worry about the dynamics of the traffic at that moment. So, when the traffic slows down in front of me, and the speed decreases below the desired speed from 100 km/h to 90, 80, 70, my car automatically keeps enough distance. And I have to put less effort and energy into driving.”* [36]. Another use of technological interventions can be to check one’s cognitive functioning.

The (technological) tools participants use seem to comply with their needs to get support regarding their complaints. An example of this is Anouk, who indicated that she has concentration issues resulting in difficulties with writing messages. She uses an AI tool to check her emails, which complies with her disabilities.

Sub-theme 8.3 – Obstructing aspects

Although many of the participants experience benefits from technology, **technology can also be experienced as obstructive**. Especially, a mismatch between the technological intervention and Long COVID is experienced as obstructing. Maaïke experienced this with her smartwatch, she explained: *“My watch only drops during the day. Even if I lie down or do exercises, I have the feeling that I somewhat relax. But sometimes it can give blue regarding stress, so I am resting, but it does not indicate that I am recharging. While it feels like a recharging moment for me.”* [37]. But also, regarding energy management and complaints, phone time, and errors in technology can be experienced as disturbing.

Lastly, Henk indicated that he is not willing to let technology determine his planning. He explained: *“I won’t let my watch dictate my life. For me, it is just to look at how the situation is. I don’t accept that it says that I slept well while still feeling tired. When I am tired, I do something else; I don’t want to be fatigued. So, I perceive it more as some piece of information saying things are not all that bad, so you can do something.”* [38]. It seems that he wants to maintain a certain sense of autonomy in which there is no place for (current) technology.

4.2.10 Theme 9 – Desires technology

Half of the people shared the explicit belief that they look forward to technology that supports them in everyday life; **they experience a desire to get support (from technology) to comply with their needs**. This emphasizes the potential of technology adoption in everyday life with Long COVID.

Some participants attribute potential to technology in supporting cognition-involved activities. Veerle shared her view regarding support in information-processing: *“If you need, for example, help with something, it would be nice to fill in what your problem is, and it will show you with your video what you should do. So ‘sit down, put your socks off’, in a step-by-step plan.”* [39].

The potential for management of complaints was also recognized by two participants explicitly. Bente remarked: *“If you think about it, you feel yourself getting tired. But you are not aware of it, not actively. But if at that moment there would be technology that would perceive this or intervene, saying ‘I notice that it is time to do this for five minutes.’”* [40].

And some participants experience that current technologies could be improved or adjusted to fit better for Long COVID users. Noortje explained: *“I think regarding Long COVID, the Fitbit can be more specific for this user group. So not focused on sporting and doing your best, but more on preventing tension.”* [41].

5. Discussion

This thesis explored how people with Long COVID manage their everyday life with a focus on cognitive challenges, and investigated the current and potential role of technology to support these. The motive was to address the relative lack of insight regarding management methods of Long COVID-related cognitive impairments on the one hand, and, on the other hand, highlight the expertise of people with Long COVID regarding their own condition.

Through three sub-questions, the main research question, *“What is the (potential) role of technology in the management of cognitive aspects of everyday lives of people with Long COVID?”* was answered. This was done using a human-centered, qualitative approach, including a sociodemographic questionnaire, an activity diary, and an in-depth semi-structured interview. This methodology aimed to minimize the impact on the participant and encourage reciprocity while maximizing the insights into daily routines, self-management strategies, coping mechanisms, and technological support. The results were nine (main) themes describing the impact of Long COVID on everyday life, coping mechanisms and (cognitive) strategies to manage associated challenges, and the use of technological tools and the potential for support.

In this section, the findings are interpreted and discussed considering the theoretical framework. This is done through emphasizing three axes among the themes, describing the relevant (underlying) relations¹⁰. After this, the research questions are answered in light of the interpretations. The third part includes methodological limitations and recommendations for future research methodologies, research, and technology design.

5.1 Managing uncertainties: People with Long COVID are expert control engineers of their everyday life

A fundamental underlying interpretation of the findings is the intangibility of management of Long COVID regarding everyday life. Participants described their condition as elusive, unpredictable, and overwhelming. Many aspects are controllable (to a certain extent) through the implementation of strategies, planning methods, and coping mechanisms. The participants indicated that they adjust their daily planning regarding, among others, their activities and length of exertion, considering their capacities and impairments. However, people with Long COVID also need to act upon the uncertainty of the condition’s manifestation and the impact of (unexpected) external factors during the day.

People with Long COVID actively manipulate the input and output of their days, coping with uncertainties and their personal context. This active role people with Long COVID take in the management of their condition complies with the influential paper of Barlow et al. (2002) on self-management methods. According to this view, self-management is the ability of people to manage their symptoms, treatment, physical and psychosocial consequences, and lifestyle changes that are involved in living with a chronic condition. In order to protect their health-related condition and their

¹⁰ It should be noted that because of the explorative nature and data-intensity of this study, there are many potential interpretations of the findings. For the sake of clarity and length of the report, the three most relevant relations are emphasized.

quality of life, people need to monitor their condition and adjust their planning regarding cognitive, physical, social, and mental aspects.

Based on the findings, people with Long COVID adhere to these self-management norms. They constantly adjust their planning based on the insights they gather through monitoring their health-related complaints and biometrics through smartwatches. This is described in, among others, subtheme 7.1. An example of this is that participants indicated they take a rest after exertion or after (unexpected) stress during the day. A direct consequence of this is that people with Long COVID are, besides their everyday planning and activities, constantly managing their (mental) expectations.

People with Long COVID take it a step further regarding their self-management as they do not only focus on their health-related condition; they also consider their mental and social desires and challenges. An example of this is that participants indicated that they sometimes deliberately cross their health-related boundaries to meet these desires; they build in pleasure and comfort. This complies with the paper by Leggat et al. (2024), which explains that people with Long COVID implement personal strategies protecting mental and health-related boundaries and needs, and adjusting expectations. In practice, they walk a very thin line between all the dimensions involved to optimize their everyday life.

This process can be seen as an extensive control loop. People with Long COVID try to optimize their day to their expectations, needs, desires, wishes, and responsibilities. They adjust their daily planning regarding expected complaints and activities, which can be defined as feedforward control. However, during the day, their health-related condition is affected by the elusive nature of Long COVID and uncontrollable events, e.g., external factors. Feedback in terms of complaints or warnings from their management devices is used to eventually adjust their planning and control their condition regarding everyday life. This means that besides the feedforward trade-off of expectations, needs, desires, wishes, and responsibilities regarding their planning, people with Long COVID are in an ongoing negotiation with themselves, deciding what is best at every moment during the day, considering every involved element.

Although participants are not always satisfied with their everyday life, and they cannot always find the optimal path, it can be concluded that they have developed some admirable self-management methods to cope with their complex situation. They are control engineers working to optimize their everyday life as best as possible through implementing an (almost) all-encompassing control system.

5.2 Extending cognition: Facilitating cognitive processes through active implementation of strategies and tools

Apart from methods to manage health-related complaints during the day, the findings show that people with Long COVID also implement strategies to facilitate cognitive activities in everyday life. The participants indicated that they adjust most cognition-involved activities during the day when possible. These strategies not only serve as a way to minimize cognitive effort and reduce the likelihood of overload but also to manage stress and respect personal limits. This perspective was, among others, identified in sub-theme 7.2.

These approaches to handling everyday activities can be understood through the lens of epistemic actions as described by Kirsch & Maglio (1994). This fundamental paper presents the concept of

reducing the cognitive load of a certain task through simplifying the associated mental computation. Through manipulating the task environment, the task is easier and thus supports cognitive functioning. This aligns with the adjustments people with Long COVID deliberately and, sometimes unconsciously, make to facilitate the cognitive process involved in a certain everyday activity. Through adapting the mental and/or physical environment of a task involving cognitive functioning, e.g., processing stimuli, they actively contribute to a reduction in cognitive load.

Restructuring the task takes place through tactics but also using (technological) tools and strategies as described by the participants. This also aligns with the concept of extended mind as described by Clark & Chalmers (1998); cognitive functioning not only takes place within the mind but can be extended by external tools and structures. An example of this is the use of lists and notes. Participants indicated that they use this as a way to support their memory and reorganize their thinking during an activity. This means that processing stimuli and memory are (partially) outsourced to tools; the cognitive tasks take (partially) place through a list and the interaction between the list and the individual. Thus, people with Long COVID actively implement strategies, tools, and technological devices to expand their cognitive capacities through extending the environment in which cognitive functioning takes place.

5.3 High potential, but the realization of cognitive support through technological tools is missing

Existing literature within the theoretical framework showed that most technology used by people with Long COVID is used for self-management regarding monitoring complaints, planning, and training cognitive functioning (AbilityNet, 2024; Katella, 2024; Orcha Health, 2021). This was also observed in the findings of this study. Participants indicated to use technology mainly to monitor complaints and eventually adjust planning or activities. For example, some actively implement feedback from, e.g., their wearables to adjust their planning or activity. This aligns with research by Khondakar & Kaushik (2023) which describes the potential of wearables as a tool in Long COVID healthcare. However, this study shows that this biometric data is mainly used by living experts themselves as input for their self-management methods. This confirms the potential of wearables in a Long COVID context and shows that it is already used and implemented by people themselves. This technological feedback and control are mostly physically based. However, because people adjust their planning and activities, it is not only a way to control physical well-being but also contributes (indirectly) to their cognitive capacities because of complaint management.

On the other hand, the findings show that the cognitive potential of technology is not yet realized. Although not widespread among participants, it was seen that they all had (even when not directly mentioned) significant difficulties with cognitive activities such as sending letters, answering messages, but also other daily processes, such as cooking or showering, can be tough. These difficulties and explicitly stated desires for support in cognitive activities, as described in theme 9, show that there is potential for technology. There seems to be a particular desire for support in processing stimuli within a task. However, only limited technological support was implemented by participants to facilitate these processes.

The current technological tools that are available do not seem to match this demand. This complies with existing literature, described in the theoretical framework, on tools regarding cognition. These mostly include cognitive training to improve one's functioning and treat brain fog as described by studies of (Florian, 2022; NHS Helath Research Authority, 2023; Research Team BEACON, 2023) or more hands-on/immediate support through time management apps, text-to-speech software, and mood monitoring tools (AbilityNet, 2024; Orcha Health, 2021). Some participants indicated that they use these as well. However, the potential for supporting processing stimuli within everyday activities remains. Despite the strategies people implemented, the difficulties with information-dense and stimuli-intense activities persist.

Based on the extended mind theory of Clark & Chalmers (1998) and Heersmink's (2015) application of this theory, it can be concluded that a cognitive artifact can be of great use. Heersmink (2015) describes how cognitive artifacts such as technological tools can serve as an extension of one's cognitive function. As processing stimuli is, despite personal strategies and self-management methods, still a large problem, there is great potential for technology to support this. Technological tools could facilitate enabling everyday activities which include cognitive functioning.

The potential of technology is also shown in the mismatch between current technology and users' needs and desires. Some participants experienced that technological tools do not fit their Long COVID-related needs. This is coherent with prior research on the use of existing technologies that are designed for other conditions or situations (Khondakar & Kaushik, 2023; Orcha Health, 2021). This research shows that this can lead to misunderstanding or misfit in human-technology interaction. However, despite this mismatch, people still use, among others, their wearables as a support to manage their condition. Thus, the benefits outweigh the disadvantages. It seems that adjustments or modules in these accepted technologies are a relatively simple way to support people with Long COVID even better; there is only limited work on work on trust and adoption.

5.4 Interpretation-based conclusions

5.4.1 Impact on everyday life

Long COVID is experienced as elusive, unpredictable, and overwhelming. As a result, it determines completely people's interpretation of everyday life. This aligns with the first sub-research question, which addresses the impact of Long COVID on everyday life: *"How does Long COVID affect the everyday lives of people who experience fatigue, cognitive dysfunction, and eventual PEM-related symptoms that severely reduce their ability to carry out day-to-day activities?"*. The answer is that Long COVID is all-encompassing. It impacts people's social, physical, mental, and cognitive functioning and activities substantially. As a result, people cannot live the life they want or be who they were before their coronavirus infection.

5.4.2 Self-management of Long COVID

Despite this impact, people with Long COVID have developed and implemented self-management methods and strategies; they take an active role in the facilitation of their everyday life. People with Long COVID are not bystanders of their condition. Instead they actively implement methods to manage the related challenges. This includes managing their health-related situation despite the uncertainties, but also adjusting activities to facilitate cognitive functioning. This is related to the

second sub-question, which focuses on the adjustments people with Long COVID make to execute and be involved in everyday life: *“How do people with Long COVID-related fatigue and cognitive dysfunction adjust and implement cognitive strategies and technology in order to manage the cognitive aspects of their everyday life?”*.

Managing their situation regarding the lens of their desires, needs, and wishes is an ongoing process; they have to constantly adjust their predicted planning, activities, and mental state during the day. As a result, they cannot rely on any certainties that facilitate ‘normal and healthy’ everyday life. This has far-reaching implications for social life, but also physical and cognitive activities, and people’s mental state; Long COVID is a constant presence that needs to be considered for every minute of the day. In practice, this means that people with Long COVID are in an ongoing negotiation with themselves regarding their health-related complaints, desires, and needs regarding activities and planning.

Despite the lack of treatment, people with Long COVID developed their own, personal self-management approaches to cope with their health-related conditions and associated uncertainties. They monitor their health-related state and condition, both themselves through complaints, and by smartwatches through biometric data. As a result, they gain insight into their condition and they adjust their planning and activities in real-time, considering expectations, needs, desires, wishes, and responsibilities accordingly.

This active role was also seen regarding cognitive functioning. People with Long COVID actively implement methods to facilitate their everyday life through extending cognitive functioning. This is done using epistemic actions, but also the use of strategies and (technological) tools which support cognitive activities that involve, e.g., working memory and processing stimuli.

Thus, these self-management methods, including strategies, coping mechanisms, and (technological) tools do not only facilitate people with Long COVID, enabling them to execute certain activities, but also give them (part of) the lead back in their own story. They are experts in self-management and actively look for solutions to respond to their challenges in everyday life.

5.4.3 The potential of technology

Although all participants have already implemented technology in their everyday lives as part of (cognitive) strategies and self-management, there is much potential for future use. This aligns with the third research question, which focuses on the potential of technology to support everyday life with Long COVID: *“What do people with Long COVID currently miss in their developed strategies and implemented technological supporting tools to carry out daily cognition-involved activities, and what can be improved?”*. There is especially a role for extending processing stimuli within cognition-involved activities. These could be implemented in current strategies and coping mechanisms. Besides, the mismatch between the needs and desires of people with Long COVID and current technologies can be solved. Adjustments or modules in accepted technologies are a relatively simple way to support people with Long COVID even better, there is only work on trust and adoption.

5.5 Recommendations for future work

5.5.1 Methodological limitations and recommendations

Although the background and the Long COVID symptoms of the participants were diverse, this study revealed some important similarities. Despite the diversity in background and health-related complaints, experiences, and strategies of people overlapped (for example, in themes 2 and 6). Besides, because of the exploratory character of this research, each participant's shared experience can be seen as valuable insight into the (potential) use of technology in everyday life with Long COVID and to generate potential ideas for future work, like technology design (recommendations). Therefore, this study recommends future research in this field to consider a similar methodological approach regarding participants.

Another discussion point regarding the background is the educational and/or job characteristics of the participants. The individuals who shared their lived experience were almost all working in mainly cognition-based functions, like a secretary or engineer, in an intellectual environment like a consultancy firm or an educational institution. However, people with more physically focused jobs are missing. Their experiences towards, among others, the planning of everyday life, fatigue, rest, and work differ from the research sample. This means that their self-management of Long COVID and associated experiences probably differ as well. As this group is a substantial part of society, it is important that they are heard as well. Although this research tried to reach a diverse group of people, this part of society is missing. Therefore, it is crucial to involve people from all socio-economic backgrounds and develop a strategy for this in future research.

Another limitation regarding the research sample is that the people who signed up to participate have noticed the recruitment message in some way and were willing to participate. This means that they are in contact with their social environment or involved in social media and are open to sharing their experiences; they seem to have an active mindset (or are able to have this mindset). However, this means that the people who are too ill to go on social media or have lost their social contacts because of their condition were not involved in this project. So, although this research aimed to make everyday life with Long COVID more visible, there remains a group of people invisible.

Furthermore, this research investigated through the experiences of people with Long COVID how they adjust their everyday life regarding their (health-related) condition using cognitive strategies and (technological) tools. This was the leading input for answering the research question. However, a result of this methodology is that crucial insights are missing towards these strategies and tools; it is difficult for people to recognize the adjustments they make because of their condition, because these are deeply embedded in their everyday life. Therefore, the participants could have seen some examples of how they adjust their everyday activities and planning. It should be noted that the activity diary already activated thinking about strategies and tools people implement in their lives to support them. Several participants indicated that filling in the activity diary led to a different view on everyday life and gave them insight into their daily routines and adjustments. However, there can still be missing crucial insights. Therefore, it would be valuable for future research to optimize this part of the research methodology towards awareness, so the full scope of tools and strategies would be captured. A suggestion for this is to visit participants at home to evaluate (potential adjustments in the) living environment and potential present (technological) tools, and observe how they manage certain daily activities.

Resignation and mental state pose an additional research methodological limitation. This could have led to missing insights into how current strategies and technologies could be improved. Participants in this research actively worked on accepting their situation and had some sense of resignation towards their condition. During the data collection, they were asked to think about the things they are missing in their current (self-)management methods. However, this goes against the well-developed mental coping mechanisms of the participants; they try to focus on the positive features of their everyday life instead of the negative elements. Thus, they protect themselves against the potential negative effects of complaining. This is an explanation for the lack of information and insights into what is missing in their current (cognitive) strategies and technological tools. This (potential) protection mechanism was not taken into account during the development of the research methodology. Although the research method was developed to be as comfortable as possible for the participants, it was not considered how to handle (mental) coping mechanisms that counteract the goal of collecting negatively loaded insights and experiences about everyday life strategies and the use of technology. This should be considered in future research.

A further methodological limitation concerns the origin of participants' (cognitive) strategies. Although in some interviews the development of strategies and implementation of (technological) tools was discussed, this research did not focus on where participants' ideas for self-management methods came from. However, this would have given a good overview of how the care system (including general practitioners, occupational therapists, and physiotherapists) and other information sources play a role in forming these strategies and implementing technological tools. This could be a good method to map the knowledge and translation gaps. Besides, these insights would have been an additional explanation for the implementation of these strategies.

5.5.2 Recommendations for research and design

Apart from the recommendations that were proposed regarding methodological limitations, there are also some other suggestions regarding future research and design projects about supporting people with Long COVID in everyday life through cognitive strategies and technology. These recommendations were composed through deductive reasoning based on (the discussion of) the results and limitations of the research.

The research field of Long COVID is young and therefore unexplored, especially regarding the topics of cognitive support and human-technology interaction, meaning that there is an infinite number of matters to investigate and design for. However, this discussion aims to end with some concrete recommendations for future research and design.

- This research focused on cognitive support through technological tools. However, theme 8.2 shows that technology can also serve as a communication provider and even as mental support through its social features. Although only a few participants mentioned this social aspect of their human-technology interaction in everyday life, this connection could (potentially) be beneficial for one's mental state. Therefore, this research suggests looking further into the (social) relation between people with Long COVID and the technological tools they use in everyday life. Besides, this finding shows that there is a potential for technology to fulfill a role as social support in the lives of people with Long COVID. This also corresponds to their desires regarding social contact, which are complicated by their (health-related) complaints.

Therefore, it is valuable for designers to look into the idea of technological interventions that can serve as a social companion that complies with the social and health-related needs of people with Long COVID.

- Another suggestion for future research regarding human-technology interaction in the context of the everyday life of people with Long COVID is the adoption of technology. On the one hand, some participants indicated a mismatch between their own experience and the conclusions of their technological tools (such as heart rate data on wearable devices), but still adjusted their behavior on the tool, while some other participants disregarded this technological input. Thus, there seems to be a skewed balance of adoption of technological advice; some people follow these insights despite the mismatch, while others want to retain their autonomy. Therefore, future research should investigate how the adoption of technological tools among people with Long COVID takes place, and how this is related to the mismatches between (subjectively) experienced (health-related) symptoms, and the technological output (which can be seen as an error). This is especially necessary because of the precarious balance people with Long COVID need to manage in their everyday life, which means that the impact of errors differs from people without Long COVID.
- The discussion of the results concluded that there was a mismatch between the mental state and the ratio of people with Long COVID regarding their identity and experience of productivity. A potential explanation of this could lie in the cultural standards that, partially, contribute to one's identity. However, the impact of culture on the mental state is beyond the scope of this study. Therefore, this study encourages researchers with a sociological, psychosocial, or anthropological background to investigate how mental state regarding Long COVID and other (semi-)chronic conditions is related to cultural standards.
- Current technological tools that play a role in the (self-)management of the everyday life of people with Long COVID are not customized for their condition. This means that they do not comply with their needs. This leads to a mismatch between the tool and the user. Therefore, future technology should be adjusted to the demands of people with Long COVID. This can imply several things:
 - Existing tools could be extended with a Long COVID-specific module. It was seen that, despite the mismatch, many participants use technologies to manage their complaints. Thus, the current potential of these devices is already optimally used by this group of users. However, to optimize their use, future design should consider their condition and the associated needs. For example, by focusing on rest instead of productivity, and implementing certain disclaimers regarding the data about, e.g., the body battery.
 - The function of existing tools should be replaced by a Long COVID-specific technological intervention. This means that a new research and design cycle should be set up in which people with Long COVID are involved from the start to develop fitting technological tools. Their expertise should be leading, as they are best aware of the needs and desires regarding (cognitive) assistance in everyday life activities.
- Designers of technological assistance tools could investigate how they can support information processing within activities. One of the participants indicated this as a desire in theme 9. As people with Long COVID mostly encounter difficulties regarding stimuli-processing in cognition-involved everyday tasks, there is great potential for technological

assistance. This could be, for example, an AI-involved system that recognizes where the user's process went wrong, advising on how to continue or solve the issue or error. In this way, the technological intervention is part of the user's cognitive process on the aspects that are not working well enough to handle these activities.

- People with Long COVID have already developed cognitive strategies. However, designers and researchers could investigate whether these cognitive strategies could be optimized through the use of technological tools.
- Long COVID is a complex condition with multiple (organ) systems within the body involved, leading to several health-related symptoms and disabilities. Therefore, the needs of people with Long COVID can be seen as an ultimate use case for accessibility in technology. So, this research encourages designers to actively involve people with Long COVID in the design process, learn from them, and use their expertise as a guideline in their design. Besides, because of this complexity and manifestation of health-related symptoms, this study and future research and (technology) design in the field of (cognitive) assistance of Long COVID in everyday life can be useful for (research about) the management of other, related, diseases and conditions such as burn-out, ME/CFS, depression, chemo brain, and the consequences of a concussion. The methodology and the researcher's attitude can serve as a useful perspective for other research on these topics.

This research is one of the first in its field. It shows that (cognitive) strategies and technological tools can support people with Long COVID in the cognitive aspects of everyday life. As long as there are no (proven) medical treatments for (the cognitive symptoms of) Long COVID, self-management methods are the only way to lighten the impact of this condition on the everyday lives of people who are living with it. Besides, there were many things found regarding cognitive strategies and use of technology (that do not per se comply with current literature) because of the expertise of people with Long COVID. It can be concluded that their input is crucial for the usability of design and truthful research. Therefore, the researchers of this project strongly encourage other researchers to work further on the topic of cognitive assistance (through technological tools) with the active involvement of people with Long COVID as experts.

6. Conclusion

Severe and life-limiting Long COVID symptoms have a major impact on the quality of life of people with Long COVID. Self-management methods are seen as the best option to manage this condition regarding complaints and everyday life. However, current self-management methods are mainly focused on physical symptoms, while cognitive functioning is crucial for daily activities. Therefore, this thesis aimed to contribute to more insight into how people with life-limiting Long COVID can be supported in the cognitive aspects of their everyday lives using a technology-focused human-centered approach, investigating the (potential) role of technology.

This was done using a participant-friendly methodology in which people with Long COVID were involved as experts. In-depth interviews, after filling in a sociodemographic questionnaire and an activity diary, in which eleven people with Long COVID shared their experiences, provided insight towards the impact of Long COVID on everyday life, the current strategies and technology that are implemented to support cognitive aspects of everyday life, and what is currently missing in their (technology-based) self-management methods.

The results show that Long COVID affects all substantial aspects of everyday life; people with Long COVID are impaired regarding everyday activities, involvement in (societal) everyday life, managing their (health-related) condition, social contact, and mental state, and are constantly looking for balance between their complaints, expectations, and personal desires. Besides, they actively look for solutions to manage everyday life regarding activities and planning towards the challenges that are caused by their condition through implementing strategies and technological tools. Lastly, it was seen that the role of technology could be taken a step further by optimizing complaint management, considering the Long COVID-specific needs. Besides, the potential of cognitive support through technology was confirmed by current use and desires regarding technology.

Thus, the answer to the research question *“What is the (potential) role of technology in the management of cognitive aspects of everyday lives of people with Long COVID?”* is that currently, technology play a role in the self-management of everyday life with Long COVID, especially regarding (complaint) management and execution of cognition-involved activities. Future technology could play a substantial role in processing stimuli and, in an optimized form, in complaint management, complying with the needs and desires of the user group, contributing to their quality of life.

This research encourages researchers and designers to actively involve people with Long COVID in future projects as experts. This study shows that they are solution-focused, have developed several strategies, and implement technological tools to support them with everyday life challenges, and they are aware of their needs and desires regarding (technological) tools.

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Appendix A: Literature

Appendix A.1: Symptom list Davis

This is (part of) the supplementary material of the paper on the symptoms of Long COVID written by Davis et al. (2021) presenting the Long COVID symptoms per category.

Symptom categories

Symptoms are grouped in 10 categories, as below, given which organ/system they manifest in: Systemic: fatigue, temperature, weakness, flushing and sweating related symptoms, and post-exertional malaise

- Neuropsychiatric - because of the number and prevalence of neuropsychiatric symptoms assessed, they are broken into the following nine sub-categories and discussed separately:
 - Cognitive Functioning
 - Memory
 - Speech and Language
 - Sensorimotor Symptoms
 - Sleep
 - Headaches
 - Emotion and Mood
 - Taste and Smell
 - Hallucinations
- Cardiovascular: heart rate, palpitations, blood pressure (excluded from in-depth analysis, see Supplemental Figure S5), visibly bulging veins, clots, and pain/burning in the chest
- Dermatologic: itchiness, rashes, and obvious changes in skin and nails
- Gastrointestinal: GI upset, hyperactive bowel sensations, and appetite-related symptoms
- Pulmonary and Respiratory - encompasses breathing, coughing, and sneezing, and oxygen saturation related symptoms (excluded from in-depth analysis, see Supplemental Figure S5)
- Head, Ear, Eye, Nose, Throat (HEENT): both physical and sensory symptoms related to the eyes, ears, nose, mouth, throat, and face including facial paralysis and numbness. Headaches are captured in the Neuropsychiatric category.
- Reproductive, Genitourinary, and Endocrine: symptoms related to menstruation and lack thereof, symptoms related to male reproductive function, symptoms related to sexual function, symptoms related to thirst and urinary function, and low and high blood sugar (excluded from in-depth analysis, see Supplemental Figure S5)
- Immunologic and Autoimmune: new and heightened immune responses
- Musculoskeletal: chest tightness and aches and pain throughout the musculoskeletal system

Symptom list

Total of eight symptoms from the following list were excluded from our analyses, as their measurement required specialized equipment or tests that many participants may not have had access to. Excluded symptoms included 1. high blood pressure, 2. low blood pressure, 3. thrombosis, 4. confirmed seizures, 5. suspected seizures, 6. low oxygen levels, 7. high blood sugar, and 8. low blood sugar.

Non-neuropsychiatric symptoms

Cardiovascular	1. Blood clots (Thrombosis) 2. Bradycardia 3. Fainting 4. Heart palpitations 5. High blood pressure (if measured)	6. Low blood pressure (if measured) 7. Pain / burning in chest 8. Tachycardia 9. Visibly inflamed / bulging veins
Dermatologic	1. Brittle or discolored nail 2. COVID toes 3. Dermatographia 4. Itchy skin 5. Itchy, other	6. Other skin / allergy symptoms 7. Peeling skin 8. Petechiae 9. Skin rashes
Gastrointestinal	1. Abdominal pain 2. Constipation 3. Diarrhea 4. Feeling full quickly when eating 5. Hyperactive bowel sensations	6. Loss of appetite 7. Gastroesophageal reflux 8. Nausea 9. Vomiting
HEENT Head, Ear, Eyes, Nose, Throat	1. Bloodshot eyes 2. Blurred vision 3. Changes in the voice 4. Changes to the ear canal 5. Double vision 6. Dry eyes 7. Ear pain 8. Eye pressure or pain 9. Facial paralysis 10. Facial numbness - left side 11. Facial numbness - other 12. Facial numbness - right side 13. Floaters 14. Hearing loss	15. Itchy eyes 16. Lump in throat 17. Numbness of ear 18. Other ear / hearing issues 19. Other eye issues 20. Pink eye (Conjunctivitis) 21. Redness, outside of eyes 22. Runny nose 23. Seeing things in peripheral vision 24. Sensitivity to light 25. Sore throat 26. Total loss of vision 27. Tunnel vision 28. Vision symptoms, overall
Immunological	1. Changes in sensitivity to medication 2. Heightened reaction to old allergies 3. New allergies	4. New anaphylaxis reaction 5. Shingles
Musculoskeletal	1. Bone ache or burning 2. Joint pain 3. Muscle aches	4. Muscle spasms 5. Stiff neck 6. Tightness of chest
Pulmonary	1. Breathing difficulty (Normal O2) 2. Cough with mucus production 3. Coughing up blood 4. Dry cough 5. Low oxygen levels (<94%)	6. Other respiratory symptoms 7. Rattling of breath 8. Shortness of breath 9. Sneezing
Reproductive	1. Abnormally heavy periods / clotting 2. Abnormally irregular periods 3. Bladder control issues 4. Decrease in size of testicles/penis 5. Early menopause 6. Extreme thirst 7. High blood sugar (if measured)	8. Low blood sugar (if measured) 9. Other menstrual issues 10. Other semen / penis / testicles issues 11. Pain in testicles 12. Post-menopausal bleeding 13. Sexual dysfunction 14. Urinary issues, other
Systemic	1. Chills / flushing / sweats 2. Coldness 3. Elevated temperature (98.8-100.4 F) 4. Fatigue 5. Fever (>100.4 F) 6. Heat intolerance 7. Low temperature	8. Night sweats 9. Other temperature problems 10. Post-exertional malaise 11. Skin sensations 12. Temperature lability 13. Weakness

Neuropsychiatric symptoms

Mood & Emotion	1. Aggression 2. Anger 3. Anxiety 4. Apathy 5. Delusions 6. Depression 7. Derealization/Depersonalization 8. Euphoria 9. Hypomania	10. Impulsivity and Disinhibition 11. Irritability 12. Mania 13. Mood Lability 14. Other Mood Symptoms 15. Sense of doom 16. Suicidality 17. Tearfulness
Cognitive Dysfunction	1. Acute confusion/disorientation 2. Agnosia 3. Cognitive dysfunction (Brain fog) 4. Difficulty problem-solving 5. Difficulty thinking	6. Difficulty with executive functioning 7. Other Cognitive Symptoms 8. Poor attention 9. Slowed thoughts 10. Thoughts moving too quickly
Hallucinations	1. Auditory Hallucinations 2. Hallucinations, other	3. Tactile (touching) Hallucinations 4. Visual (seeing) Hallucinations
Headaches	1. Headaches, at the base of the skull 2. Headaches, behind the eyes 3. Headaches, diffuse (entire brain) 4. Headaches, in the temples	5. Headaches, other 6. Headaches after mental exertion 7. Migraines
Memory Issues	1. Forgetting how to do routine tasks 2. Inability to make new memories 3. Long-term memory loss	4. Other Memory Symptoms 5. Short-term memory loss
Sensorimotor	1. Dizziness / vertigo 2. Electrical zaps 3. Inability to cry 4. Inability to yawn 5. Neuralgia (nerve pain) 6. Numbness/loss of sensation 7. Numbness on one side of the body 8. Seizures (confirmed)	9. Seizures (suspected) 10. Sensation of brain pressure 11. Sensation of brain warmth/on fire 12. Sensitivity to noise 13. Tingling/prickling/pins and needles 14. Tinnitus 15. Tremors 16. Vibrating sensations
Sleep	1. Awakened feeling unable to breathe 2. Difficulty falling asleep 3. Insomnia 4. Lucid dreams 5. Nightmares 6. Restless leg syndrome	7. Sleep apnea 8. Vivid dreams 9. Waking up early in the morning 10. Waking up during the night
Language & Speech	1. Changes to non-primary language 2. Difficulty communicating in writing 3. Difficulty communicating verbally 4. Difficulty comprehending others 5. Difficulty finding the right words	6. Difficulty reading 7. Difficulty speaking 8. Other speech / language symptoms 9. Slurring words / speech 10. Speaking unrecognizable words
Smell & Taste	1. Altered sense of smell 2. Altered sense of taste 3. Heightened sense of smell 4. Heightened sense of taste	5. Loss of smell 6. Loss of taste 7. Phantom smells 8. Phantom taste

Appendix B: Methodology

Appendix B.1: Positionality statement

As the first and primary researcher of this thesis, I am aware that my experiences, sociodemographic and medical background, and (cultural) biases could impact the research process and outcomes. Therefore, these are actively considered during the research process through the principle of reflexivity.

Position regarding the group of interest

A prominent aspect to consider is my insider position in the group of interest during this thesis, I have had Long COVID for more than two (and a half) years. This manifested mainly in 1) fatigue in social, cognitive, and physical aspects; 2) cognitive dysfunction; 3) physical complaints like difficulties in condition, and pain in the head, heart, and muscles; and 4) Post-Exertional Malaise. My condition had been slowly recovering from severe Long COVID, which was life-limiting, to milder symptoms. This illness and the recovery process were still going on during this research project.

This personal, direct involvement has both advantages and disadvantages. My insider position contributes to trust, openness, the feeling of being heard and accepted, and empowerment among the participants. However, this could also be a confusing role as I am a participant as well; there could exist a conflict between my own experience and the lived experience of participants. Another potential disadvantage is that participants assume that I understand them and am aware of certain underlying principles, which leads to less information being provided. Besides, because of the lived experience, I could be led by empathy and understanding, which could affect my critical thinking.

Sociodemographic background

Apart from my position within the population of interest, my sociodemographic background could potentially affect the research process. The most prominent factor is my culture, in which opinions about recovery and responsibility are seriously intertwined; I grew up in a neo-liberal Western system during a period in which self-sufficiency was heavily endorsed by politics and society. This concept implies that people are owners of their weaknesses and are responsible for 'repairing' these to comply with a certain norm. Although personally, I do not adhere to this concept, I am still sensitive to this kind of thinking because of its role in my culture. This could impact my thinking about how people with Long COVID act upon their impairments and the way they look at (the recovery of) their condition. As I am aware of this risk, during the data collection and analysis, I constantly evaluated my thinking critically.

Some other sociodemographic characteristics that could influence the research process are my sex and gender, age, and education. I identify myself as a woman, which could positively affect the participants' feeling of connection, as many of the participants are female; this is an additional insider's position. Besides, age could also influence the perception of the participants; I am in my mid-twenties, meaning that I am relatively young and thus relatively inexperienced. This could change participants' attitude towards the research's quality, but also regarding my knowledge. Another characteristic is my education level. My academic background can be positive as this could serve as an indication of the importance of researching this topic, and also give me, as the first researcher, the capacity to critically reflect on and recognize patterns in the data. However, it could

also be negative as it could lead to a lack of common ground with participants who have different backgrounds, and it could lead to skepticism regarding participants' viewpoints when these differ from academic sources. Therefore, during the data collection and analysis, I focused on experiences and differentiated between one's opinion and experience.

Reflexivity during the data collection and analysis

During the project, I have taken several measures to minimize potential biases. The most important one is the active involvement of other researchers in the methodology development, data collection, and data analysis; they critically reflect on my input and check the data analysis. Other measures I have implemented are thorough literature research to develop the research method and include multiple points of view, keeping a research diary, reflecting on every interview afterward, and critically evaluating potential biases I have. This is an ongoing process that has led to, among others, new perspectives, adjusted approaches to interviewing and interpreting data, and changes in thinking and actions.

Below are a few examples of this reflexivity during the research process:

- One of the participants indicated that (s)he followed a certain therapy, which I am skeptical about. I recognized my resistance and discussed this with the second researchers. I changed my attitude to curiosity and focused on how (s)he experienced everyday life with Long COVID during the interview and the data analysis.
- I noticed an additional insider position during conducting the interviews: most of the participants (and the Long COVID population) were women, and I am a female researcher. Several participants seem to be hypersensitive to this element and emphasized this similarity. This could have provided confidence to share additional information or experiences they thought were female-related matters, like psychological issues and social constructs. During the data analysis, this was considered critically. In the following interviews, I did not expand on these matters from the viewpoint of our shared characteristic.
- During the interviews and phone calls with participants, I experienced how hard it could be at some points to separate my own experiences of living with Long COVID from my researcher's role. This research is personal as you work together with people with Long COVID and investigate their experiences with the condition. So, I want to give them a good feeling by confirming their answers and connecting through shared experiences. After a few interviews, I prepared a few sentences for myself to sustain the researcher role while still maintaining my empathetic attitude.
- Some of the participants compared the current (societal) situation with that before, from their view of living with Long COVID. They explained that things were going differently when they were young than it is currently going on. Emphasizing this could be declared by the age differences between us; in these cases, my life experience seemed to positively influence participants' information sharing.
- During the definition of the themes in the data analysis, I noticed having difficulties with reporting certain negative-loaded matters, such as the impact of Long COVID on the lives of people, including a negative self-image. I wanted to keep a positive perspective and report this. Besides, by documenting the negative feelings of participants towards themselves and their identity, I would share intimate and complicated information. I discussed this matter

with one of the second researchers, and we came up with a strategy in which we used my own lived experience as a guideline. For the findings in the intimate and complicated themes, I asked myself how it would feel for me if these would be shared, and based on this, I would conclude whether and how to report on it. So, my empathy and ethical concerns towards the participants got in the way, but my lived experience could serve as a solution.

- During the data analysis and documentation, I observed that I was projecting my perspective on society towards the explanation for the negative feelings towards one mental state. Although I did not immerse myself in the topic of culture and societal norms and values on mental self-image, I still developed a certain view on these interconnections through information and opinions from newspapers and podcasts. I recognized that this was not suitable for this project and therefore focused on the facts, without discussing the findings from a colored viewpoint. As the relation between culture and identity could influence the (mental) impact of Long COVID, this is briefly noted in the recommendations for future work section in the discussion, and suggested to do future research about this possible relation.

To conclude, I am aware of my socio-demographic background and insider position regarding the group of interest. These have affected the research methodology, data collection, and data analysis. The factors that were within my reach have been adjusted to improve the quality of the research. During the data collection and analysis, I constantly evaluated the process and did active introspection. Based on this reflection, adjustments regarding methods or findings, or ideas were implemented. This contributed to the transparency and quality of this thesis.

Appendix B.2: Descriptions of the participants

In this section, the people who participated in this thesis were described. Based on data from the sociodemographic questionnaire, the first contact during the briefing, the activity diaries, and the interviews, an overview of each participant was created to get an insight into their background. The names are randomly generated and thus do not correspond with the actual names of the participants.

Anouk

Anouk is a woman in her early fifties who got COVID-19 in February 2024. She suffers from fatigue, concentration problems, memory problems, shortness of breath, hypersensitivity to light or sound, muscle weakness, PEM (post-exertional malaise), nausea/reflux, loss of fitness, and dizziness. Before developing Long COVID, she was diagnosed with allergies, allergic asthma, a leaking heart valve, irritable bowel syndrome, and osteoporosis.

Due to Long COVID, Anouk had to quit her job. Currently, she is working on a therapeutic basis and coaches a colleague of hers. She partially lives with her young adult son, who helps her with heavy and practical tasks. When he is around, she is more likely to get overstimulated because of his presence and interaction. Besides, her sister and brother assist her with household-related chores, and her friends walk her dog and drive her around. During this research project, she was starting with so-called 'cogmed' training, which is an intervention program to enhance working memory and concentration.

Bente

Bente is a woman in her late twenties with a background in human-technology interaction who got COVID-19 in March 2023. She suffers from fatigue, concentration problems, memory problems, and headaches. During her illness process, she recovered partially, but a few months before this research project, she had a relapse, which forced her to start again with recovery. At the moment of her participation, she had already taken major steps in recovery.

Despite this recovery, Bente is not able to perform her former job and has a different function within the same company. Her family and boyfriend support her, but she has lost many friends because of her condition. Her boyfriend does many household-related chores, but she also tries to help him with that, even though she does not have the energy. They have got a dog which helps her with her planning and physical activities; this leads to a feeling of rest.

Daphne

Daphne is a woman in her early fifties with a job as secretary of a consultancy company. She got COVID-19 in December 2020 and since then has suffered from fatigue, concentration problems, memory problems, hypersensitivity to light and sound, headaches, and loss of fitness. During this process, she partially recovered but also got a few new coronavirus infections, resulting in relapses.

Due to Long COVID, Daphne works less, and the settings have changed into a better-fitting situation. She is living alone and has a LAT relationship. Her family, husband, and some friends support her. However, her social life has changed a lot as she has lost touch with a large part of her social network. Since she is doing better, she tries to participate again sometimes in social activities, although this does not always suit her condition.

Henk

Henk is a man in his early sixties who lives in Germany. He got COVID-19 in May 2022 and, since then, has suffered from fatigue, concentration problems, memory problems, muscle pain, muscle weakness, sleeping problems, and pain in the arms or legs. During this process, he experienced ups and downs regarding his complaints. Before he developed Long COVID, the participant suffered from high blood pressure and tinnitus. At the time of this project, Henk had contact with a neurologist and psychologist about his Long COVID and sometimes followed a German healthcare program for people with Long COVID.

Henk is not able to work anymore because of his Long COVID. Because of his age, he does not have any financial issues and thus financial-related worries. He lives with his wife in a spacious apartment directly next to a park and the mountains. He experiences that this has helped him during his (recovery) process. His wife and children support him where needed.

Karin

Karin is a woman in her early sixties. She got COVID-19 in January 2023 and since then has suffered from fatigue, concentration problems, hypersensitivity to light and sound, muscle weakness, loss of fitness, problems with sleeping, shortness of breath, heart palpitations, dizziness, fatigue after, buzzing in the ears, hypersensitivity to television, mobile phone, crowds, and conversations. Despite her complaints, no medical causes were found in the regular medicine. Therefore, she looked into alternatives and currently follows the mind and body vision (TMS).

Because of her condition, Karin could not work for almost two years, but the UWV only defined her as partially disabled (for only 5%). Therefore, she had to start working again and currently works at a library within set restrictions. She lives in a peaceful neighborhood together with her two adult children and husband. Because of problems within her relationship, recovery is tough. In online sessions about the mind and body vision, she finds (mental) support.

Lotte

Lotte is a woman in her mid-fifties who was a secretary at an educational institution. In October 2022, she got COVID-19 and, after recovery from this initial illness, she turned back to her former everyday life for a month, but her complaints increased over time, resulting in decreased abilities. In the end, she had to stop working in March 2023 because of Long COVID. Since then, she has suffered from fatigue, concentration problems, hypersensitivity to light and sounds, muscle weakness, loss of fitness, problems sleeping, dizziness, PEM (post-exertional malaise), and pain in the arms and legs. In May 2024, she started taking antidepressants to control her fears and panicking feelings, decrease the intensity of her emotions, and get her nervous system to rest, which helps her.

Lotte lives with her husband in a peaceful neighborhood, and because of her husband's job, she is alone during the day on business days. This rest is good for her and her condition. Her family and friends understand her situation and support her. At the moment of the research project, she followed ReAttach therapy by her occupational therapist.

Maaïke

Maaïke is a woman in her early sixties. In October 2022, she got COVID-19. She developed vague complaints that increased over time. In May 2024, she stopped working, and in July 2024, she was diagnosed with Long COVID. She suffers from fatigue, concentration problems, shortness of breath, hypersensitivity to light and sound, headaches, muscle pain, loss of fitness, dizziness, pain in arms and legs, and PEM (post-exertional malaise). Before her diagnosis, she was also diagnosed with a shortage of vitamin B12.

Maaïke lives together with her husband in a peaceful and quiet environment. As it is a ground-floor home, she does not need to climb the stairs. She experiences her living environment as helpful in her life with Long COVID. She also has adult children who visit her, and she visits them. At the moment of the research project, she followed a treatment trajectory by an occupational therapist and physiotherapist.

Noortje

Noortje is a woman in her mid-thirties with a background in mental health care as a caregiver. She has two young children for whom she has cared for the last few years instead of working. In March 2022, she got ill with the COVID-19 virus, resulting in long-term complaints including fatigue, concentration problems, memory problems, hypersensitivity to light and sounds, headaches, dizziness, muscle weakness, loss of fitness, and PEM (post-exertional malaise). Besides, she suffers from migraines, especially caused by hormones.

Noortje is not able to work and has lost a lot of social contacts because of her condition. She lives with her children and her husband and handles most of the household chores. During the day, the neighborhood is quiet, but the summer evenings can include a lot of disturbing sounds. She sometimes sees a friend or plays board games with friends. During the study, she followed physical strength training at a physiotherapy practice and a lifestyle program.

Roos

Roos is a woman in her mid-fifties. She got COVID-19 in April 2022 and developed concentration problems, memory problems, hypersensitivity to light and sound, headaches, dizziness, difficulties with finding the right words, difficulties with her short-term memory, difficulties learning new things, can only execute one task at the time, inability to handle stress/pressure, low cognitive capacities, and difficulties with reading. During her condition's process, she was diagnosed with hypothyroidism since the COVID-19 infection, and for this, she was treated with medication, resulting in the recovery of physical complaints.

Because of her Long COVID, Roos is not able to execute her former job. Because of her experience within the organization and resulting trust, she now coaches her colleagues and works during adjusted times. She lives with two young adult sons and her husband. They support her with practical matters, together with a friend. Her house lies within a quiet, green neighborhood close to nature, which she has experienced as helpful for her (recovery) process.

Veerle

Veerle is a woman in her early forties. She has lived in Spain for two years with her husband and two young children. She got COVID-19 in December 2023, resulting in long-term complaints including fatigue, concentration problems, hypersensitivity to light and sound, loss of fitness, and dizziness.

Because of her Long COVID, Veerle had to quit her job. Also, household chores cost her a lot of energy, so her husband took over a lot of tasks, including children-related tasks. Sometimes, friends support them with their family logistics.

She lives in a house with a garden near the beach, which helps her to get out of her house despite her restricted abilities. Her children take a lot of her limited energy, but she values the presence of this commotion.

Wout

Wout is a man in his mid-sixties who worked as an engineer. He got COVID-19 in December 2020 and, since then, experienced multiple relapses because of new infections and the impact of vaccinations. He experiences concentration problems, memory problems, hypersensitivity to light and sounds, headaches, and PEM (post-exertional malaise). Before Long COVID, he had diabetes, joint problems, and problems with sleeping.

Because of his medical issues in combination with his age, Wout has quit his job. Because of a beneficial financial regulation, he does not experience worries about his financial situation. He lives together with his wife and one of his adult children in a quiet neighborhood with little to no sound inconvenience because of traffic or other people, and nearby nature. He experiences this as beneficial for his Long COVID (recovery) process, as this supplies his need to get rest and take a walk or ride the bike in the woods. His family and friends support him during his illness.

Appendix C: Materials

Appendix C.1: Recruitment material

Recruitment text

Gezocht: deelnemers voor onderzoek naar cognitieve ondersteuning bij Long COVID

Ben jij of ken jij iemand met Long COVID met klachten zoals vermoeidheid en cognitieve dysfunctie heeft? Dan ben ik op zoek naar jou!

Ik zoek mensen die:

- Al minstens een zes (6) maanden Long COVID hebben (een officiële diagnose van een arts is **niet** nodig);
- Klachten ervaren zoals vermoeidheid, cognitieve problemen (denk aan moeite met taal, concentratieproblemen, vergeetachtigheid of hersenmist) en eventueel PEM (Post-Exertional Malaise);
- Door Long COVID beperkingen in hun dagelijks leven ervaren en aanpassingen hebben moeten doen;
- Nederlands spreken;
- Tussen de achttien (18) en zevenenzestig (67) jaar oud zijn.

Mijn naam is Emma en voor mijn masterscriptie van Human-Technology Interaction aan de TU/e, doe ik onderzoek naar de (mogelijke) rol van technologie in het ondersteunen van de cognitieve aspecten in het dagelijks leven van mensen met Long COVID. Als ervaringsdeskundige met Long COVID wil ik met dit onderzoek bijdragen aan oplossing om met een ziekte en zoals Long COVID om te kunnen gaan. Hiervoor wil ik samenwerken met de ware experts: mensen met Long COVID zelf. Er is momenteel nog maar weinig (betrouwbare) informatie over dit onderwerp, dus help je mee?

De studie bestaat uit:

- 1) Een korte vragenlijst;
- 2) Het bijhouden van een dagboek voor een paar dagen;
- 3) En een (online) interview.

Het onderzoek vindt plaats in januari, februari en maart en wordt waar mogelijk aangepast naar de behoeften, wensen en uitdagingen die bij Long COVID komen kijken. Daarnaast sta ik open voor aanpassingen zodat je zo comfortabel mogelijk kunt deelnemen aan dit onderzoek.

Wil je meedoen of meer weten? Stuur mij dan een DM via LinkedIn (Emma Höngens) of een mail (e.hongens@student.tue.nl).

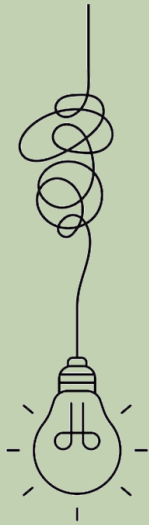
Voldoe je zelf niet aan de criteria voor mijn onderzoeksgroep, maar ken jij iemand die mogelijk geïnteresseerd is? Stuur dit bericht dan gerust door. Alvast hartelijk bedankt!

Emma Höngens

Master student Human-Technology Interaction, Technische Universiteit Eindhoven

e.hongens@student.tue.nl

Recruitment flyer



Gezocht: Long COVID experts

Mensen die minstens een half jaar Long COVID hebben en last hebben van vermoeidheid en cognitieve klachten

Waarvoor?

Een onderzoek naar de (mogelijke) rol van technologie in het ondersteunen van de cognitieve aspecten van het dagelijks leven van mensen met Long COVID

Emma Höngens
Student Human-Technology Interaction
e.hongens@student.tue.nl

Appendix C.2: Briefing and debriefing

Briefing – phone

Hallo [naam],

Je spreekt met Emma Höngens van het onderzoek naar de rol van technologische ondersteuning van cognitieve aspecten van het dagelijks leven van mensen met Long COVID. Bel ik gelegen?

Zoals ik al kort aangaf in de mail van [dag invullen] wil ik graag met je doornemen hoe het onderzoek eruit ziet. Ik wil je eerst wat achtergrondinformatie geven over waarom ik dit onderzoek doe en daarna wat ik van jou verwacht en wat je van mij kan verwachten.

{Waarom dit onderzoek?}

Zoals je waarschijnlijk weet, is er nog relatief weinig (betrouwbare) informatie over Long COVID en hoe je hiermee kunt omgaan. Uit onderzoek dat tot nu toe is uitgevoerd, blijkt dat vooral 'zelfmanagement' effectief kan zijn het controleren van Long COVID-klachten en mogelijk herstel.

Deze onderzoeken en methodes focussen zich vooral op de ondersteuning en controle van fysieke klachten, terwijl de cognitieve aspecten minstens zo belangrijk zijn – en misschien belangrijker – voor het uitvoeren van dagelijkse activiteiten. Met dit onderzoek wil ik in kaart brengen hoe mensen met Long COVID momenteel omgaan met het cognitieve aspecten van het dagelijks leven en wat er ontbreekt in de ondersteuning om activiteiten met een cognitieve belasting uit te voeren. Dit doe ik met een technische bril op; technologie kan goed worden ingezet om mensen te ondersteunen in allerlei dagelijkse taken.

In dit onderzoek kijken we dan ook naar welke rol technologie nu speelt bij mensen met Long COVID in het dagelijks leven en waar technologie potentieel een rol kan spelen om mensen te ondersteunen bij het uitvoeren van dagelijkse activiteiten die cognitieve belasting vragen.

Veel onderzoek maakt gebruik van kennis van ergotherapeuten, artsen en fysiotherapeuten om de onderzoeksvraag te beantwoorden. Dit onderzoek vraagt om een andere aanpak; samen met de echte experts, namelijk mensen met Long COVID zelf, wil ik werken aan een antwoord op de vraag wat de (mogelijke) rol van technologie is in de ondersteuning van de cognitieve aspecten van het dagelijks leven met Long COVID. Jouw ervaringen en inzichten als ervaringsdeskundige spelen hierin een grote rol.

Wat denk je hiervan? Heb je hier vragen over?

{Wat houdt het onderzoek in?}

Het onderzoek bestaat uit drie onderdelen:

- 1) Een korte vragenlijst die je online invult;
- 2) Een activiteitendagboekje dat je invult voor een paar dagen (dit kan op papier of op een tablet, wat jij fijn vindt);
- 3) En een interview over hoe je omgaat met cognitieve uitdagingen en welke rol technologie speelt met behulp van je ingevulde dagboekje. Dit kan online of op de TU/e campus, wat jij prettiger vindt.

Mijn doel is dat jij je in dit onderzoek gehoord voelt. Ik heb zelf ook al langere tijd Long COVID en weet dus hoe lastig sommige dingen kunnen zijn. Daarom heb ik het onderzoek zo veel mogelijk aangepast om het makkelijker te maken om mee te doen. Als je graag wil meedoen, maar dat niet lukt door je Long COVID-klachten, dan denk ik graag met je mee over aanpassingen aan het onderzoek zodat het wel bij jou situatie past. Het kan nog steeds zo zijn dat het voor jou te veel is, dus ik vind het belangrijk dat je weet dat je dat op elk moment kan aangeven. Verder mag je altijd om hulp vragen. Dus als je iets niet goed begrijpt, maar ook als je moeite hebt met begrijpen wat ergens staat of het lezen van een document niet lukt.

Is het allemaal duidelijk tot nu toe?

{Afspraken maken}

Ik zou graag nog een paar dingen met je afspreken om wat dingen te plannen en het onderzoek zo soepel mogelijk voor jou te laten verlopen. Heb je daar nog energie voor momenteel?

Zoals ik je net kort vertelde is het tweede onderdeel van dit onderzoek het invullen van een activiteitendagboekje. Hierin vraag ik je om een creatief formulier in te vullen voor een week. Wil je deze graag op papier invullen of op je tablet, als je die hebt?

Als je het op papier wil invullen, sturen wij je een boekje op via de post dat je kan invullen. Hiervoor hebben we een adres nodig. Waar kunnen wij het naartoe sturen? (Dit kan ook nog in de mail)

(opmerking voor onderzoeker: adres herhalen)

Als tweede zou ik graag afspraken willen maken voor het interview. Ik weet dat zo'n gesprek vaak veel energie kan vragen van iemand met Long COVID, dus ik vind het belangrijk dat we dat zo prettig mogelijk voor je maken. Geef vooral aan wat het fijnste voor jou is. Dit interview doen we op het einde van het onderzoek en dan kijken we naar je activiteitendagboekje, stel ik wat vragen en luister ik naar jouw ervaringen en ideeën. Dit duurt zo'n 30-45 minuten in totaal, afhankelijk van hoeveel jij te delen hebt. Dan weet je een beetje hoe het eruit ziet.

- We kunnen het interview online doen of op de campus van de TU/e in Eindhoven afnemen. Wat is voor jou fijner, om het interview online of offline te doen?
- Wat voor software gebruik jij meestal voor videobellen? /Wat is handig voor jou?
- Welke dag zou de komende weken voor jou uitkomen? Tien dagen van tevoren vraag ik je te starten met het activiteitendagboek.
- Tijdens het interview wil ik tussendoor pauzes inlassen om de inspanning te verdelen en zo de impact te verkleinen. Daarom wil ik graag weten hoe lang jij je kan concentreren achter elkaar en hoe lang een pauze voor jou moet zijn om het te laten werken.
- Heb je hier eventueel nog ideeën over?
- Zijn er andere dingen die het makkelijker maken voor jou om het interview te doen?

Dankjewel.

Zijn er verder dingen jij nodig hebt zodat je mee kan doen aan het onderzoek? We kunnen eventueel kijken naar aanpassingen als dat het voor jou makkelijker maakt.

Heb je nog vragen?

{Wat moet je nu doen?}

Ik ga je na dit telefoongesprek alles wat ik zojuist heb verteld, nogmaals via de mail versturen zodat je dat op je gemak nog eens door kan lezen. Verder heb ik een ingevuld toestemmingsformulier van je nodig voordat je mee mag doen aan het onderzoek. Het is belangrijk dat je deze goed doorleest of laat doorlezen door iemand in je omgeving en het daarna ingevuld terugstuurt naar mijn email. Als je merkt dat dit niet lukt of als je het fijn vindt hierbij hulp te krijgen, kan ik dit formulier ook met je doornemen.

Als het formulier is getekend, kan het onderzoek echt beginnen en stuur ik je de link van de vragenlijst en het activiteitendagboek.

Is het allemaal duidelijk voor je?

Heb je na dit gesprek vragen of opmerkingen over het formulier of iets anders? Je kan mij altijd mailen, bellen of een berichtje sturen. Ik ben meestal beschikbaar op werkdagen tussen 09.30-16.30, maar dit kan af en toe variëren door mijn Long COVID-klachten.

Heel erg bedankt voor je tijd en energie. En ik spreek je snel weer. Dag!

Debriefing after interview

Heel erg bedankt dat je jouw ervaringen en inzichten over het dagelijks leven met Long COVID en (mogelijk) gebruik van technologie ter ondersteuning van cognitieve taken hebt gedeeld met mij. Het was fijn om met je samen te werken. Ik waardeer het ontzettend dat je je beperkte energie hebt gebruikt om mij te helpen mijn afstudeeronderzoek.

Als je je de komende weken nog dingen bedenkt die mogelijk relevant zijn voor mijn onderzoek, dan mag je deze altijd mailen. Alle input telt!

De komende tijd ga ik de informatie en ervaringen die je met mij hebt gedeeld samen met wat andere deelnemers aan dit onderzoek hebben gedeeld analyseren. Ik hoop hiermee mijn onderzoeksvragen te kunnen beantwoorden en op die manier meer inzicht in hoe technologie kan worden gebruikt om mensen met Long COVID zoals jij en ik te ondersteunen in hun dagelijks leven. Als je wil, kan ik mijn uiteindelijke bevindingen met je delen via de mail of je bellen. Daarnaast kan ik je ook mijn scriptie sturen als deze is gepubliceerd.

Appendix C.3: Consent form

Informatieblad voor proefpersonen

Dit formulier geeft je informatie over het onderzoek 'de (potentiële) rol van technologie in het ondersteunen van de cognitieve aspecten van het dagelijks leven van mensen met Long COVID'.

Meedoen aan dit onderzoeksproject is jouw keuze. Je deelname aan dit onderzoek is vrijwillig en brengt geen lichamelijke, juridische of financiële risico's met zich mee. Je hoeft geen vragen te beantwoorden die je niet wilt beantwoorden. Je deelname is vrijwillig. Dit betekent dat je jouw deelname op elk gewenst moment mag stoppen door dit te melden bij de onderzoeker. Je hoeft niet uit te leggen waarom je wil stoppen met deelname aan het onderzoek. Ook kan je nog achteraf (binnen 24 uur) weigeren dat je gegevens voor het onderzoek mogen worden gebruikt. Het stopzetten van deelname heeft geen nadelige gevolgen voor jou.

Tijdens je deelname aan deze studie kunnen je vragen worden gesteld die je als (zeer) persoonlijk kunt ervaren, vanwege de gevoelige aard van het onderwerp. Deze vragen gaan met name over onderwerpen die betrekking hebben op de impact van Long COVID op je dagelijkse leven zoals vragen over je geestelijke gezondheid en de verschillen met je leven voordat je Long COVID kreeg.

Voordat je besluit tot deelname, willen wij je vragen de volgende informatie door te lezen, zodat je weet waar het onderzoek over gaat, wat er van je verwacht wordt en hoe wij omgaan met je persoonsgegevens. Op basis van die informatie kan je middels de toestemmingsverklaring aangeven of je toestemt met deelname aan het onderzoek.

Je bent natuurlijk altijd vrij om vragen te stellen aan de onderzoeker via e.hongens@student.tue.nl. Ook kun je de informatie uit dit informatieblad bespreken met mensen die je vertrouwt.

Wie zijn wij?

Het onderzoek wordt uitgevoerd door Emma Höngens, masterstudent, onder supervisie van Prof. dr. Wijnand IJsselsteijn, ir. Ans Tummers-Heemels en E. Bartholomeus van de Human-Technology Interaction groep van de Technische Universiteit Eindhoven.

Participanten

Je kan meedoen aan dit onderzoek omdat je meer dan een half jaar Long COVID hebt met cognitieve klachten en vermoeidheidsklachten en daardoor beperkt bent in je dagelijks leven, volwassen bent (tussen de 18 en 67 jaar oud) en Nederlands spreekt.

Doel en nut van het onderzoek

Het doel van dit onderzoek is om een completer beeld te krijgen van de impact van Long COVID op het dagelijks leven, de manieren waarop mensen met Long COVID hun cognitieve uitdagingen aanpakken en het huidige en potentiële gebruik van technologie ter ondersteuning van hun dagelijks leven. De verkregen informatie wordt gebruikt om in kaart te brengen wat de huidige rol van technologie is in de ondersteuning van de cognitieve aspecten van het dagelijks leven van mensen met Long COVID die cognitieve klachten en vermoeidheidsklachten hebben en waar potentie zit om deze ondersteuning uit te breiden en/of te verbeteren. Deze verkregen informatie kan worden gebruikt in toekomstig onderzoek naar ondersteuning van mensen met Long COVID om zo de kwaliteit van hun dagelijks leven te bevorderen

Wat houdt deelname aan de studie in?

Je neemt deel aan een onderzoek waarbij we informatie zullen verzamelen door:

- Jou te vragen een korte online vragenlijst in te vullen over de achtergrond en eigenschappen van jou. Dit is nodig om de onderzoekspopulatie te omschrijven voor toekomstig onderzoek en als contextinformatie voor de onderzoekers om jou beter te begrijpen;
- Jou te vragen een activiteitendagboekje bij te houden voor zeven dagen. Het doel is om hiermee inzicht te krijgen in hoe je omgaat met je energie en je cognitieve klachten gedurende de dag. Deze kan je digitaal of fysiek invullen, afhankelijk van jouw voorkeur.
- Jou te interviewen over de rol van jouw (persoonlijke) strategieën en eventueel gebruik van technologie om om te gaan met de cognitieve aspecten van jouw dagelijks leven met Long COVID. Hierbij gebruiken we de informatie van de vragenlijst voor de context en het door jou ingevulde activiteitendagboekje. Het interview vindt online plaats of op de campus, afhankelijk van jouw voorkeur. Je antwoorden worden opgenomen via een audio- en/of video-opname. Er zal ook een transcript worden uitgewerkt van het interview.

Welke persoonsgegevens verzamelen en verwerken wij van jou?

Wij verzamelen de volgende persoonsgegevens die noodzakelijk zijn voor het doel van het onderzoek:

Categorie	Persoonsgegevens	Doeleinde	Bewaartermijn
<i>Contactinformatie</i>	<i>Naam, email, adres, woonplaats, telefoonnummer</i>	<i>Met jou communiceren gedurende het onderzoek, (eventueel) bevindingen delen na het onderzoek</i>	<i>5 maanden na gegevensverzameling</i>
<i>Audio- en video-opnames</i>	<i>Audio- en video-opnames</i>	<i>Interview uitvoeren</i>	<i>5 maanden na gegevensverzameling</i>
<i>Medische gegevens en contextinformatie</i>	<i>Long COVID klachten, medische diagnose(s), leeftijd, geslacht</i>	<i>Achtergrondinformatie om de onderzoeksgroep te omschrijven en te gebruiken in de interviews</i>	<i>10 jaar na gegevensverzameling</i>

Je persoonsgegevens worden enkel bewaard voor de termijn zoals genoemd in de tabel hierboven. Het bewaren van je gegevens helpt ons om aan onze wetenschappelijke verplichtingen te voldoen.

Na deze termijn worden je persoonsgegevens verwijderd of geanonimiseerd, zodat deze niet meer aan u gelinkt kunnen worden. Tenzij je expliciet toestemming geeft voor het gebruiken van herkenbare gegevens over u (bijvoorbeeld als u wilt dat uw naam of uw quotes worden vermeld in publicaties).

Als je besluit je deelname te stoppen, dan gebruiken wij je persoonsgegevens vanaf dat moment niet meer in het onderzoek.

Met zorgen of vragen over de omgang met persoonsgegevens kunt u een mail sturen naar de Functionaris Gegevensbescherming dataprotectionofficer@tue.nl. Ook kunt u een klacht indienen bij de Autoriteit Persoonsgegevens.

Vertrouwelijkheid en gebruik, opslag en delen van data

Er zullen audio- en video-opnames gemaakt worden en contactinformatie gedeeld die u kunnen identificeren, maar deze zullen gecodeerd worden en alleen met de betrokken onderzoekers van dit project worden gedeeld.

Het doel van het verzamelen, analyseren en opslaan van deze data is om de onderzoeksvraag te beantwoorden en de resultaten te publiceren in wetenschappelijke literatuur. Om uw privacy te beschermen zullen alle persoonlijke data die u kunnen identificeren opgeslagen worden op een versleutelde server van de Human-Technology Interaction groep die alleen toegankelijk is voor geselecteerde stafleden van de HTI-groep voor ten minste 10 jaar. Geen informatie die gebruikt kan worden om u persoonlijk te identificeren zal met anderen worden gedeeld.

De verzamelde data kunnen ook relevant blijken voor toekomstige onderzoeksprojecten binnen de Human-Technology Interaction groep. De gecodeerde data die in deze studie verzameld zijn en die gedeeld wordt met anderen zal (voor zover wij weten en kunnen vastleggen) geen informatie bevatten die u kan identificeren. Daarin zullen de antwoorden staan die u geeft gedurende de studie, inclusief demografische en relevante medische gegevens (zoals leeftijd, geslacht en diagnoses naast Long COVID) als u besluit die data te delen tijdens deze studie.

Bij alle onderzoeken van Human-Technology Interaction wordt gewerkt volgens de ethische code van het NIP (Nederlands Instituut voor Psychologen). Deze studie is goedgekeurd door de Ethische Commissie van de HTI-groep van de Technische Universiteit Eindhoven.

Nadere inlichtingen

Als je vragen hebt, indien je jouw deelname wil stopzetten of je nog verdere informatie zou willen hebben over dit onderzoek, dan kan je je wenden tot Emma Höngens, email; e.hongens@student.tue.nl.

Voor eventuele klachten over dit onderzoek kan je terecht bij Wijnand IJsselsteijn, email; w.a.ijsselsteijn@tue.nl of Ans Tummers-Heemels, email; a.i.m.tummers-heemels@tue.nl.

Je kunt onregelmatigheden op het gebied van wetenschappelijke integriteit rapporteren bij vertrouwenspersonen van de TU/e.

Geïnformeerde toestemming

De (potentiële) rol van technologie in het ondersteunen van de cognitieve aspecten van het dagelijks leven van mensen met Long COVID

1. Ik heb voldoende informatie over het onderzoeksproject uit het losse informatieblad gehaald. Ik heb het informatieblad gelezen en heb daarna de kans gehad om vragen te stellen. De vragen zijn naar tevredenheid beantwoord.
2. Ik weet dat mijn deelname volledig vrijwillig is. Ik weet dat ik kan weigeren deel te nemen en dat ik mijn deelname op elk moment tijdens de studie kan stopzetten, zonder opgave van redenen. Ik weet dat ik de toestemming om mijn gegevens te gebruiken kan intrekken tot 24 uur nadat de gegevens zijn vastgelegd.
3. Ik ga ermee akkoord om vrijwillig deel te nemen aan dit onderzoek uitgevoerd door de onderzoeksgroep Human-Technology Interaction van de Technische Universiteit Eindhoven.
4. Ik weet dat mijn persoonsgegevens worden verzameld en gebruikt voor het onderzoeksproject, zoals is uitgelegd in bijgevoegd informatieblad.

Daarnaast geef ik de onderzoeker toestemming voor de volgende onderdelen van het onderzoek:

5. Ik geef de onderzoeker toestemming om de antwoorden of citaten die ik gaf tijdens het onderzoek, te gebruiken in publicaties en/of presentaties – zonder daarbij mijn naam te vermelden.

JA ☐ NEE ☐

Naam Deelnemer:

Handtekening:

Datum:

Appendix C.4: Sociodemographic questionnaire

Met deze vragenlijst willen we inzicht krijgen in de achtergrond en eigenschappen van jou en de andere deelnemers. Dit helpt ons als onderzoekers om een beter beeld te krijgen van jouw situatie en jou beter te begrijpen. Dit helpt de interviews soepeler te laten verlopen. Daarnaast zijn deze vragen belangrijk voor toekomstig onderzoek. Omdat dit een van de eerste studies is in zijn soort, is het essentieel om goed te beschrijven wie hun expertise hebben gedeeld om inzicht in het onderwerp te geven.

Het is belangrijk dat je weet dat je antwoorden uiteindelijk geanonimiseerd worden op een manier waarop ze niet meer naar jou te herleiden zijn.

With this questionnaire, we aim to gain insight into the background and characteristics of you and the other participants. This will help us as researchers to better understand your situation. This will contribute to smoother interviews. Besides, these questions are important for future research. As this is one of the first studies of its kind, it is essential to describe who shared their expertise in order to provide insight into this topic.

It is important that you are aware of the fact that your answers will be anonymized in such a manner that they cannot be tracked back to you.

1. **Naam:** Name:
Vul hier je naam (voornaam, achternaam) in: Enter your answer here
2. **Hoe beschrijf je jezelf?** How would you describe yourself?
Kies één van de volgende antwoorden: Choose one of the following answers:
 - ☐ **Man** Male
 - ☐ **Vrouw** Female
 - ☐ **Anders** Different
 - ☐ **Zeg ik liever niet** I prefer not to say
3. **Wat is je leeftijd?** What is your age?
Vul hier je antwoord in: Enter your answer here:
4. **Sinds wanneer heb je Long COVID?** Since when do you have Long COVID?
Vul hier je antwoord in: Enter your answer here:
5. **Wat is je opleidingsniveau?** What is your level of education?
Kies één van de volgende antwoorden: Choose one of the following answers:
 - ☐ **Geen opleiding** No education
 - ☐ **Basisschool** Primary school
 - ☐ **Vmbo** Preparatory secondary vocational education
 - ☐ **Havo** Senior general secondary education
 - ☐ **Vwo** Pre-university education
 - ☐ **Mbo-opleiding** Secondary vocational education
 - ☐ **Hbo-opleiding** Higher professional education
 - ☐ **Bachelor universiteit** Bachelor's degree university
 - ☐ **Master universiteit** Master's degree university
 - ☐ **Anders namelijk:** Other, namely:
6. **Wat zijn jouw Long COVID symptomen?** What are your Long COVID symptoms?

Kies één of meer van de volgende antwoorden en vul eventueel aan: Choose one or more of the following answers and add if necessary:

- ☐ **Vermoeidheid** Fatigue
 - ☐ **Concentratieproblemen** Concentration problems
 - ☐ **Geheugenproblemen** Memory problems
 - ☐ **Kortademigheid** Shortness of breath
 - ☐ **Overgevoeligheid voor licht of geluid** Hypersensitivity for light or sound
 - ☐ **Hoofdpijn** Headache
 - ☐ **Somberheid/prikkelbaarheid** Dreariness/irritability
 - ☐ **Spierpijn** Muscle pain
 - ☐ **Spierzwakte** Muscle weakness
 - ☐ **Conditieverlies** Loss of condition
 - ☐ **Hartkloppingen** Heart palpitations
 - ☐ **Slaapproblemen** Sleeping problems
 - ☐ **Duizeligheid** Dizziness
 - ☐ **PEM (post-exertional malaise)** PEM (post-exertional malaise)
 - ☐ **Pijn in armen/benen** Pain in arms/legs
 - ☐ **Anders, namelijk:** Other, namely:
7. **Heb je buiten Long COVID andere medische diagnoses (zoals diabetes, ADHD, chronische pijn, allergie of depressie)?** Do you have other medical diagnoses (like diabetes, ADHD, chronic pain, allergies or depression) than Long COVID?

Kies één of meerdere van de volgende antwoorden en vul aan waar nodig: Choose one or more of the following answers and add if necessary:

- ☐ **Ja, ik had voordat ik Long COVID kreeg al een andere diagnose** Yes, before I got Long COVID I was diagnosed with another condition
 - ☐ **Ja, ik heb tijdens mijn Long COVID een andere medische diagnose gekregen** Yes, during my Long COVID I was diagnosed with another medical condition
 - ☐ **Nee** No
 - ☐ **Anders, namelijk:** Other, namely:
8. **Zo ja, welke diagnose was dit?** If yes, which diagnosis?
Het is belangrijk dat je weet dat je dit niet hoeft te delen met ons. Hier kan je je antwoord invullen: It is important that you are aware that you don't have to share this with us. Enter your answer here:
9. **Wat deed je voor werk of studie voordat je Long COVID kreeg?** What work or study were you doing before you got Long COVID?
Vul hier je antwoord in: Enter your answer here:
10. **Is dit verandert door Long COVID?** Has this been changed because of Long COVID?

Kies één of meer van de volgende antwoorden en vul aan waar nodig: Choose one or more of the following answers and add if necessary:

- ☐ **Ja, ik ben gestopt met mijn werken/studeren** Yes, I have stopped working/studying
- ☐ **Ja, ik werk/studeer minder** Yes, I work/study less
- ☐ **Ja, ik ben van werk/studie veranderd** Yes, I have switched from work/study
- ☐ **Nee, ik doe nog steeds hetzelfde** No, I still do the same as before
- ☐ **Anders, namelijk:** Other, namely:

11. **Hoe ziet jouw sociale netwerk eruit met betrekking tot ondersteuning voor je Long COVID en de hulp die je nodig hebt?** How does your social support system look like with respect to your Long COVID and the help you need?

Vul hier je antwoord in: Enter your answer here:

12. **Zou je een beschrijving kunnen geven van je huidige woonsituatie (bijvoorbeeld met wie je woont, het type woning en de omgeving) en hoe dit (het ziekteverloop van) je Long COVID bepaald.** Could you please describe your current living situation (e.g., who you live with, housing type, and environment) and the influence on (the process of) your Long COVID.

Vul hier je antwoord in: Enter your answer here:

13. **Is er nog iets anders dat je graag wilt delen dat niet aan bod is gekomen in bovenstaande vragen?** Is there anything else you want to share that was not discussed in the above questions?

Vul hier je antwoord in: Enter your answer here:

Appendix C.5: Activity diary

Appendix C.5.1: Activity diary guidelines

Beste

Dit dagboek is het tweede onderdeel van het onderzoek naar de (mogelijke) rol van technologie in het ondersteunen van de cognitieve onderdelen van het leven van mensen met Long COVID. Het doel is om beter te begrijpen je omgaat met je energie en je cognitieve klachten gedurende de dag. En vooral, welke rol spelen jouw (persoonlijke) aanpak en eventueel gebruik van technologie daarbij, om met deze situatie om te gaan.

Het dagboek bestaat uit verschillende elementen:

- 1) **DagDiagram:** Een invuloefening om de ontwikkeling van vermoeidheid, cognitieve klachten (zoals hersenmist, moeilijkheden met taal en vergeetachtigheid) en stemming gedurende de dag bij te houden.
- 2) **Dagelijkse activiteiten:** Een evaluatie van dagelijkse activiteiten die cognitieve inspanning vereisen.
- 3) **Gebruik van technologie en strategieën:** Het in kaart brengen van gebruik van technologie en strategieën om cognitieve taken te vereenvoudigen, minder intensief te maken of om je energie gedurende de dag zo goed mogelijk te controleren. Voorbeelden hiervan zijn de volgorde van een taak aanpassen, een app die helpt met welke stappen je moet nemen binnen een activiteit of een smartwatch die aangeeft wanneer je rust kan nemen.
- 4) **DagEvaluatie:** Een paar korte vragen om te reflecteren op de dag.

Het dagboek helpt om beter te begrijpen hoe jij dingen aanpakt en technologie gebruikt om je cognitieve belasting te ondersteunen. Daarnaast geeft het mij ook waardevolle informatie voor de voorbereiding van ons persoonlijke interview. Tijdens het interview bespreken we jouw ervaringen met Long COVID in het dagelijks leven en hoe technologie je daarbij helpt of zou kunnen helpen.

In dit onderzoek zijn wij vooral geïnteresseerd in cognitieve klachten en cognitieve onderdelen van het dagelijks leven. 'Cognitief' omschrijft alles waarbij je je hersens moet gebruiken, dus denken, weten en leren. Met 'cognitieve activiteiten' worden dus activiteiten bedoeld waarbij je na moet denken en/of waarbij je hoofd prikkels moet verwerken. Met 'cognitieve klachten' worden symptomen bedoeld die bepalen hoe goed je hersenen werken, dus klachten die het moeilijker maken om te denken, concentreren of dingen te onthouden.

Op de volgende bladzijde vertel ik je hoe je het dagboek kan invullen.

Uitleg dagboek

Ik vraag je dit dagboekje een week lang in te vullen.

Wat heb je nodig voor om het dagboek in te vullen?

- Een pen (meegeleverd);
- (Markeer)stiften of gekleurde pennen (meegeleverd);
- Eventueel stickers of andere toevoegingen;
- En het dagboek zelf (meegeleverd).

Stappenplan

1.

Datum en dag invullen

Begin met het invullen van de datum en de dag.

2.

Invullen van het DagDiagram

Vul het DagDiagram in voor vermoeidheid, cognitieve klachten en stemming.

- **Vermoeidheid**
Teken hoe je vermoeidheid zich ontwikkelde van vandaag. Gebruik hiervoor een staafdiagram (één (1) staaf = één (1) uur). Geef in de legenda aan welke kleur je hiervoor hebt gebruikt.
- **Cognitieve klachten**
Teken hoeveel last je had van cognitieve klachten zoals vergeetachtigheid, concentratieproblemen, moeite met taal, etc. Gebruik een lijn in een andere kleur dan bij vermoeidheid en geef deze aan in de legenda.
- **Stemming**
Geef aan hoe je gevoel was door de dag heen. Gebruik hiervoor een lijn in een andere kleur dan die bij vermoeidheid en cognitieve klachten. Geef ook hiervoor aan in de legenda welke kleur je hiervoor hebt gebruikt.
- **Extra**
Buiten deze drie dingen kan je er voor kiezen ook nog iets anders bij te houden wat voor jou belangrijk is zoals kortademigheid of hoofdpijn.

3. Intekenen van activiteiten

Houd minstens twee activiteiten bij die energie van je hersenen nodig hadden zoals koken, naar de sportschool gaan, sociaal contact of naar muziek luisteren. Alles waarbij je merkt dat je hersens vermoeid raken en minder goed gaan werken valt hieronder.

- **Labelen**
Schrijf op een label welke activiteit je hebt gedaan en hoelang je hierover deed.
- **Tijd**
Verbind dit label met het tijdstip in het DagDiagram op de tijd dat deze activiteit plaatsvond. Dit kan bijvoorbeeld door een pijl te tekenen.
- **Beoordeling**
Geef in de vakjes op het label aan of de volgende uitspraken kloppen voor de activiteit (bijv. door een krulletje of vinkje te gebruiken):
A: De activiteit ging goed.
L: Ik heb de activiteit aangepast naar mijn Long COVID.
T: Ik heb technologie gebruikt.
Op het formulier vind je de afkortingen nog een keer.
- **Opmerkingen**
Op de lijntjes van het label kan je bijzonderheden, dingen die je opvallend vindt of extra uitleg opschrijven over die activiteit. Deze kan je ook op een andere manier delen (zie tips).

4. Aanvulling op het DagDiagram

Breid je diagram eventueel uit door belangrijke of relevante momenten in je dag aan te geven. Dit kan helpen om beter te begrijpen hoe dagelijkse activiteiten en je symptomen samenhangen. Je kan hierbij denken aan:

- Gebruik van technologie
- Prikkelers zoals cognitieve inspanning, fysieke inspanning, geluid, licht en sociaal contact;
- Fijne of vervelende momenten;
- Rustmomenten.

Je kunt ook andere dingen noteren die je nuttig vindt.

5. DagEvaluatie invullen

Als laatste stap kan je reflecteren op de dag door de DagEvaluatie in te vullen. Als je schrijven niet fijn of (te) lastig vindt, kijk bij de tips voor andere manieren/oplossingen.

Op de volgende bladzijde vind je een lijst met tips voor het invullen van het dagboek.

Tips

- Je kan het dagboek aan het einde van de dag invullen, maar je kan dit ook door de dag heen doen op bepaalde momenten. Het kan helpen om hierbij een wekker te zetten op drie verschillende moment tijdens de dag, bijvoorbeeld om 12.00 uur, 15.30 uur en om 19.00 uur.
- Er zijn meerdere onderdelen die vragen om tekst. Deze hoeft je niet op te schrijven als je merkt dat het lastig is om met tekst om te gaan of als je dit geen fijne manier van jezelf uiten vindt. Je kan ook gebruik maken van steekwoorden, tekeningen, kleuren of andere creatieve oplossingen. Of je stuurt een spraakmemo via WhatsApp om de vragen en stappen te beantwoorden en uit te voeren. Het gaat erom wat voor jou werkt en waar jij je prettig bij voelt.
- Het dagboek geeft handvatten hoe je het kan invullen, maar het is helemaal aan jou hoe je het in wil vullen, wat je wil delen en wat je zelf relevant vindt voor je eigen inzichten en het onderzoek. Dit betekent dat als je dingen niet wil invullen of juist dingen wil toevoegen dat helemaal aan jou is. Verder maakt het niet uit hoe je het invult. Als je kleuren of stickers wil gebruiken om dingen te verduidelijken of gevoelens te uiten, kan dat. En als je het fijner vindt om alleen de richtlijnen te volgen, is dat ook helemaal goed.
- Het kan zijn dat je een minder goede dag hebt en het dus lastig kan zijn om het activiteitendagboek in te vullen. Je hoeft dat dan ook zeker niet te doen. Kijk wat voor jou het meest relevant is om in te vullen op dat moment en de rest kan je dan laten.
- Je hebt in totaal veertien (14) formulieren gehad, dus meer dan nodig zijn voor de zeven (7) dagen van dit onderzoek. Op deze manier kan je het formulier opnieuw invullen als jij dat een keer graag wil. Ook kan je na deze studie verder gaan met het bijhouden van je dagen op deze manier als je dat een fijne manier vindt.

Op de volgende bladzijde vind je een voorbeeld van hoe een dagboek ingevuld kan worden. Dit is puur om je een idee te geven hoe het eruit kan zien, dus je hoeft het niet na te doen. Maak van je dagboek wat je er zelf van wilt maken.

Voorbeeld

1.

Datum en dag invullen

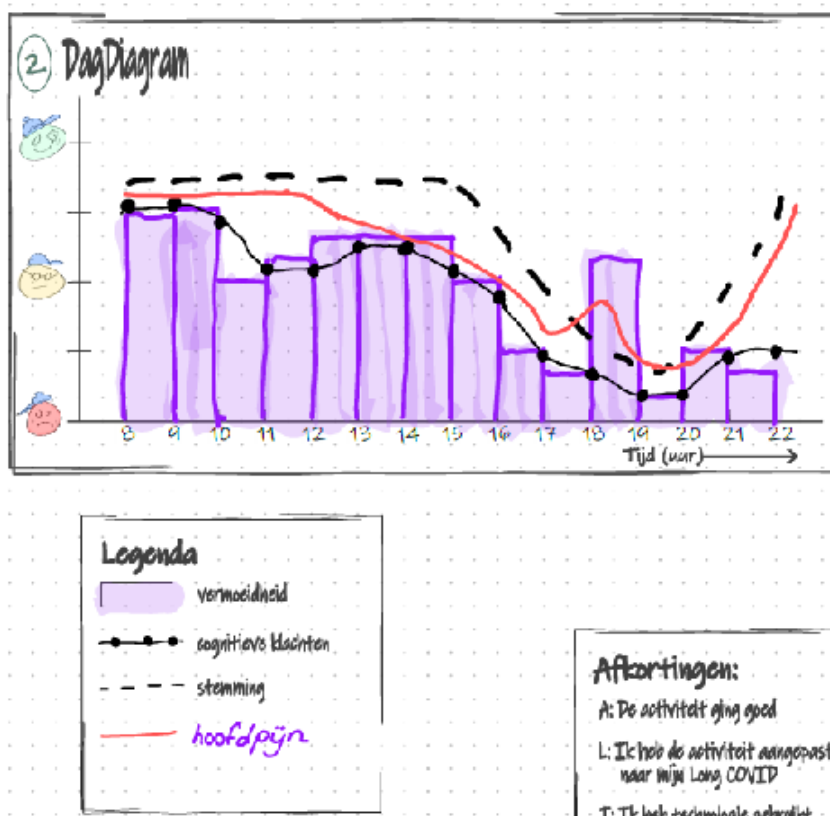
T:

1. Datum: 19 / 11 / 2024

(M) (D) (W) (D) (V) (Z) (Z)

2.

Invullen van het DagDiagram



3. Intekenen van activiteiten

3. Studeren

A: ☐ L: ☒ T: ☒

Tijd: 75 min

Gebuur chatGPT want
moete met woorden vinden
extra opmerkingen waar
moete met verwoorden
toegevoegd aan document

3. Koken

A: ☒ L: ☒ T: ☐

Tijd: 45 min

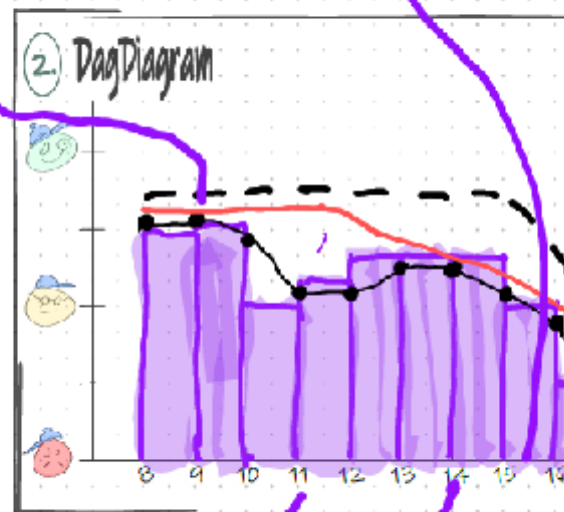
Eerst alles klaar gelegd
en gegeten, daarna
na koken + rust
moment tussendoor

3. Videobellen studie

A: ☒ L: ☒ T: ☒

Tijd: 60 min

gebruik van
notities (anders
vergeten) tussen-
door vragen om
herhaling

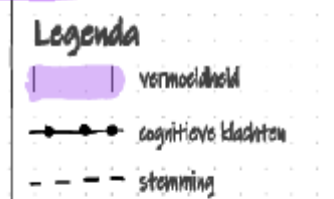


3. Wandelen

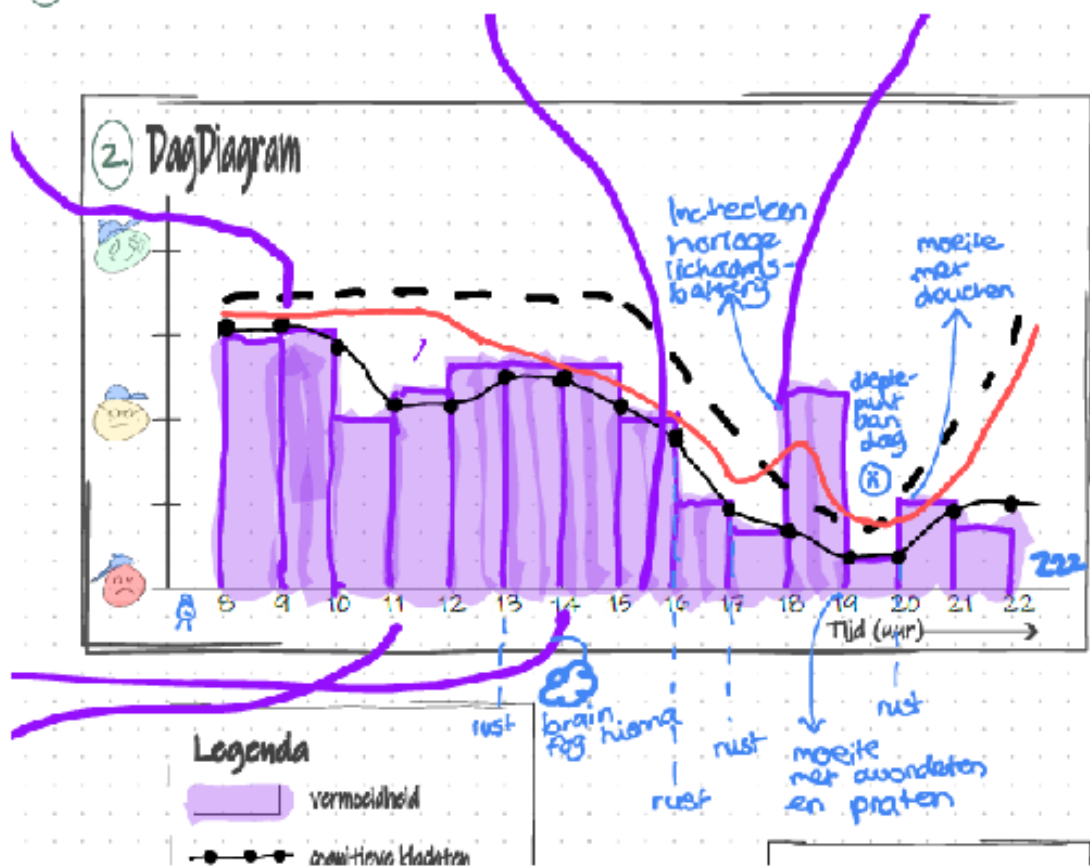
A: ☒ L: ☒ T: ☒

Tijd: 45 min

Sporthorloge gebruikt
om hartslag te
controleren voor



4. Aanvulling op het DagDiagram



5. DagEvaluatie invullen

5. **DagEvaluatie** Cijfer van de dag: 7

Dit ging goed vandaag:

Op tijd hulpmiddelen/hulp-
stra begrepen gezocht om
dingen makkelijker te maken

Veel gedaan, activiteiten
goed verspreid en
opgesplitst

Dit kon beter vandaag:

Beter opletten hoe ik me
voel + meer rustmomenten
om (cognitieve) taken uit te
voeren

Eerder/actiever opstaan

Eetmomenten beter plannen
zodat minder energie verspillen

Dit wil ik nog knijft:

Het was een productieve dag,
maar door laat avondeten
veel energie verspild en dus
sneller overprikkeld dus avond
vermoeiender dan anders

Eindresultaat

[illegible]

Appendix C.5.2: Activity diary form

1. Datum: 1 / 1

M D W D V Z

3. A: ☐ L: ☐ T: ☐

Tijd: _____

3. A: ☐ L: ☐ T: ☐

Tijd: _____

3. A: ☐ L: ☐ T: ☐

Tijd: _____

2. DagDiagram

3. A: ☐ L: ☐ T: ☐

Tijd: _____

3. A: ☐ L: ☐ T: ☐

Tijd: _____

5. DagEvaluatie

Cijfer van de dag:

Dit ging goed vandaag:

Dit kon beter vandaag:

Dit wil ik nog knijft:

Legenda

☐ Vermoeidheid

☐ Cognitieve klachten

☐ Stemming

Afsluiting:

A: De activiteit ging goed

L: Ik heb de activiteit aangepast naar mijn Long COVID

T: Ik heb technologie gebruikt

Appendix C.6: Interview schedule

Briefing interview

Hoi, welkom.

{Zorgen dat instellingen goed zijn}

We gaan zo ons interview doen. Het is geen standaard interview, maar wat opener dan in de meeste onderzoeken. Ik ben benieuwd naar wat jij te vertellen hebt en welke inzichten jij hebt. Ik stel wat vragen en ik zal ook af en toe doorvragen of terugkomen op iets wat je eerder hebt verteld, maar voor de rest ligt de regie ook bij jou in handen.

{Inchecken of er dingen zijn om rekening mee te houden zoals gillende kinderen, vertraging van de verbinding of moeite met informatie verwerken}

Zijn er verder dingen waar ik rekening mee moet houden tijdens het interview?

{Planning interview}

We hebben in het telefoongesprek afgesproken om *het gesprek op te delen in stukken van ... met ... minuten pauze tussendoor/te kijken of het lukt om het gesprek in één keer te doen*. Hoe denk je daar nu over?

{Bewust maken van rechten en stoppen}

Verder wil ik nog tegen je zeggen dat je op elk moment mag stoppen en dat je ook geen antwoord hoeft te geven op de vragen als je dit niet wil. Dit stond ook in het toestemmingsformulier wat je aan het begin van het onderzoek hebt ondertekend, maar ik benadruk het hier nog een keer omdat het zo belangrijk is. Dus als iets niet fijn vindt of als je je ergens vervelend bij voelt, laat het weten dan gaan we gewoon door naar het volgende onderwerp. En als je merkt dat je te uitgeput of moe bent om door te gaan of als je gewoon wil stoppen, dan is dat ook goed. We kunnen eventueel het interview verdelen over meerdere momenten als je merkt dat het te veel energie kost om in één keer te doen.

Als laatste is het belangrijk dat je weet dat alles wat jij zegt goed is; ik vind jouw input heel erg waardevol, want jij bent een expert op dit onderwerp en jouw ervaringen en deelname zorgen ervoor dat ik mijn onderzoek kan doen.

{Start opname}

Om het gesprek goed te kunnen analyseren maak ik een opname. Als je het goed vindt en van de voren geen vragen hebt, zou ik graag de opname willen starten. Is dat oké voor jou?

Interviewvragen

{Om welke vragen te beantwoorden, verzamelen de interviews informatie?}

- Hoe heeft Long COVID het dagelijks leven van mensen beïnvloedt die klachten hebben die het lastiger maken om dagelijkse activiteiten uit te voeren?
- Hoe passen mensen met Long COVID cognitieve strategieën en technologiegebruik aan of passen deze juist toe om de cognitieve aspecten van het dagelijks leven te ondersteunen/controleren?

- Wat mist er momenteel volgens mensen met Long COVID momenteel in de strategieën en technologie die voor handen is om dagelijkse activiteiten die vragen om cognitieve capaciteiten uit te voeren in combinatie met hun ziekte?

De **dikgedrukte** vragen zijn leidend. Het doel is deze te beantwoorden. Dit betekent dat als ik hier op een andere manier antwoord op krijg, dat ook goed is. De vragen die niet dikgedrukt zijn, zijn mogelijke manieren om door te vragen. Daarnaast vind je op het einde nog wat extra vragen die handig kunnen zijn om bij de hand te hebben.

{Introductievragen}

- **Hoe gaat het met je?**
 - o Heb je een goede of minder goede dag vandaag?
- **Hoe ging het invullen van het activiteitendagboekje?**
 - o Vond je het fijn of juist vervelend?
 - o Hoe heb je het ervaren?
 - o Heb je al eens eerder zoiets gedaan?
- Zijn er dingen die je graag wil bespreken vandaag?

{Bespreken activiteitendagboekje}

- **Zou je me willen uitleggen hoe je het activiteitendagboekje hebt ingevuld?**
 - o Wat vond je belangrijk?
 - o Waar heb je op gefocust?
- **Wat viel je op?**
 - o Welke lessen neem je mee?
- Op welke manier heb je je dagelijks leven aangepast door Long COVID?
 - o Voer je activiteiten anders uit dan voordat je Long COVID kreeg?
- Waaraan merk je dat je moe, uitgeput of overprikkeld bent?
 - o Hoe merk je dat in wat je doet?
 - o Waar loop je op zo'n moment tegenaan?
- **Hoe pak je een activiteit aan waarbij je moet nadenken als je merkt dat je hersens minder goed werken (bijvoorbeeld door vermoeidheid of overprikkeling)?**
 - o Pas je je activiteiten aan als je moe bent en/of cognitieve klachten hebt? Hoe ziet dat eruit?
 - o Pas je de structuur aan van de activiteit?
 - o Hoe maken deze aanpassingen activiteiten makkelijker?
 - o Gebruik je technologie?
- **Hoe gebruik je momenteel technologie?**
- **Wat mis je in de technologie die je gebruikt of waarvan je het bestaan weet om het mogelijk te maken dagelijkse taken uit te voeren waarbij je moet nadenken in combinatie met je Long COVID?**
 - o Wat mis je momenteel in ondersteuning buiten de technieken die je zelf toepast als je moe/(over)belast bent bij het uitvoeren van dagelijkse taken?
 - o Wat zou er verbeterd kunnen worden volgens jou in de huidige strategieën en technologische hulpmiddelen om dagelijkse activiteiten behapbaar te maken en dus haalbaarder voor jou?

{Mogelijke (extra vragen)}

- Hoe zou je zelf de samenhang van vermoeidheid, cognitieve klachten, stemming en eventuele andere klachten beschrijven? Welke verbanden vielen jezelf op?
- Welke activiteiten waarin je cognitieve prikkels ervaart, vragen de meeste energie van je?
 - o Wat zijn prikkels die dingen moeilijk maken
 - o Waar loop je op zo'n moment tegenaan?
- Noem een paar voorbeelden van activiteiten waarin participant vermoeid was of relatief cognitief beperkt
 - o Pas je je activiteiten aan als je moe bent en/of cognitieve klachten hebt?
 - o Hoe ziet dat eruit?
 - o Hoe maken deze aanpassingen activiteiten makkelijker?
 - o Gebruik je technologie?
- Hoe gebruik je momenteel technologie?
 - o Vooral ter ondersteuning van je dagelijks leven/communicatie of ook om Long COVID-klachten te reguleren/controleren?
 - o Hoe gebruik je het om ervoor te zorgen dat je dagelijkse activiteiten die cognitieve energie vragen uit kan voeren?
 - o Is je technologiegebruik veranderd ten opzichte van voor je Long COVID? Gebruik je het meer of minder?
- Hoe ziet jouw wens-ondersteunende-technologie eruitzien?

{Vragen of participant nog wat wil delen}

Is er nog iets anders dat je graag wilt delen dat niet aan bod is gekomen tijdens ons gesprek?

Debriefing interview en onderzoek

Ik wil je heel erg bedanken voor dit interview. Ik heb veel aan de informatie en ervaringen die je hebt gedeeld.

{Mogelijkheid bieden om extra dingen te delen naderhand}

Als je de komende tijd nog iets bedenkt, dan mag je dit altijd delen met mij via de mail of de telefoon. Het kan behoorlijk lastig zijn om in één uur alles te zeggen en te delen wat nuttig is voor dit onderzoek en wat je graag kwijt wil. Daarom hoor ik graag van je als je je nog dingen bedenkt!

{Uitleggen wat er nu gaat gebeuren}

De komende weken ga ik de informatie en ervaringen die je met mij hebt gedeeld samen met wat andere deelnemers aan dit onderzoek hebben gedeeld onderzoeken. Ik hoop hiermee mijn onderzoeksvragen te kunnen beantwoorden en op die manier meer inzicht in hoe technologie kan worden gebruikt om mensen met Long COVID zoals jij en ik te ondersteunen in hun dagelijks leven. Als je wil, kan ik mijn bevindingen met je delen via de mail of je bellen. Daarnaast kan ik je ook mijn scriptie sturen als deze af is.

Voor nu wil ik je heel erg bedanken dat je je beperkte energie hebt gebruikt om mee te werken aan mijn afstudeeronderzoek. Dat betekent veel voor mij. Ik hoop dat je een positieve, betere tijd tegemoet gaat met je Long COVID.

Appendix E: Data analysis

Appendix E.1: Whiteboard codes and themes



Appendix E.2: Original interview quotes

	Original quote
1	Dus dat is wel wat het interessant maakt om daar dan even bewust bij stil te staan, van oké dat zijn wel echt de dingen die energie vreten, zeg maar. En dat laat me ook wel nadenken van oké, hoe kan ik voorkomen dat ik zo hard leegloop op die dingen?
2	Dat vond ik meer confronterend dan ik had verwacht. En ja dat je dus inderdaad toch meer aanpassingen doet omdat je ze eigenlijk automatisch doet inmiddels, en dat vind ik dan wel weer confronterend, want ik heb zelf ook vaak het idee van 'Ach, weet je, het gaat eigenlijk hartstikke goed en ja, ik kan weer heel veel en ik doe weer heel veel en ik ben bij heel veel dingen bij.
3	Ja je kunt niet het leven leiden, wat je wil. Ik kan niet aan het werk, ik kan niet naar verjaardagen, ik kan geen drukke dingen doen.
4	Is het gewoon heel gek, want je denkt douchen, wat is er moeilijk aan? Maar ik vind het anders schijnbaar lastig. Er zitten heel veel stapjes aan, je moet naar de badkamer, dan moet je je uitkleden in een bepaalde volgorde. Ja, je moet onder de douche. Water moet dan op een bepaalde temperatuur, dan moet je nadenken over oké ik moet mijn haar wassen. Shampoo dit, afdrogen, je kleren weer aantrekken.
5	Ja ik, want ik heb wel meer moeite met ook in een mailtje met de dingen verwoorden, zeg maar. Dan heb ik een telefoontje tussendoor gehad en dan had ik eigenlijk een zin in mijn hoofd. Ja, die is dan natuurlijk weer weg, die ben ik dan weer kwijt.
6	Dat vind ik nog steeds wel als ik echt in de verdrukking kom, dat vind ik echt heel vervelend. En als ik dan in die verdrukking naar binnen moet zou moeten en dan moet ik dus op dat moment ook nog gaan bedenken waar ik wil gaan zitten. Dat gaat niet goed dan, dan krijg ik inderdaad een of andere error. Dat vraagt heel veel te veel van mij.
7	Als je in de auto zit en je moet nog zeker een half uur rijden, maar je moet plassen. Ook dan krijg je druk, stress in je lichaam, dus het zit in al die dingen die tot onrust in je lijf leiden die die zorgen ervoor dat je meteen dat je eerder moe wordt.
8	Maar als ik cognitief overbelast ben, dan merk je dat ook wel aan je lijf.
9	Ja, want die maandag, toen ben ik echt wel heel hard over mijn grens gegaan. Vooral die gesprekken over mijn zoontje, dat was wel heel erg nodig, maar de dag daarna voelde ik me echt niet goed. Echt niet goed. En woensdag was het helemaal erg. Toen heb ik ook afspraken moeten afzeggen en toen heb ik de hele dag in mijn bed gelegen.
10	Maar ik denk wel die cognitieve, het heeft wel als ik heel weinig energie heb, gaat dat cognitief ook minder goed. Dat merk ik ook met bijvoorbeeld uit je woorden komen, een gesprek hebben of zo of waar ik veel last van heb is bijvoorbeeld geluid.
11	Ik heb niets geleerd, maar het is voor mij gewoon heel erg een herkenkend patroon wat steeds terugkomt. Het gaat goed, dus een paar dingen om het beter te maken wat positief is en dan heb je elke dag één actie die veel kracht kost en dan ga je berg af en wordt alles slechter.
12	Ja en soms voelen dingen ook wel anders als dat ze uiteindelijk zijn, dan heb je toch het idee dat je meer energie hebt dan dat je uiteindelijk eigenlijk hebt, zeg maar.
13	Ja niemand heeft het meer in de gaten. Ik kan wel nog steeds heel erg geraakt worden 'huh? Corona bestaat dat nog dan?'. Ja die vind ik lastig. En ook mensen die ja 'ik ga wel weer gewoon werken, hoor, want ik heb nou lang genoeg thuis gezeten'. Ja, ja, dat vind ik lastig.
14	Of dat als mensen bij elkaar zijn dan ja, dan ben ik echt daarna helemaal gesloopt.
15	Dan heb ik het gewoon daarnaast liggen dus dat doen we dan vaak samen. We hebben een heel mooi kookeiland. Dus vaak staat de één aan de ene kant en de ander aan de andere kant. Nou en dan snij ik en dan is mijn man aan het koken, zo van dat in de gaten en dan zit ik alles te snijden. Dus dat is een beetje hoe we het nu veel meer doen, deden we anders ook weleens maar nu...

16	Als ik echt een slechte dag heb, dan kan ik echt balen, Maar ik krijg heel veel energie van mensen die langskomen of van mijn zoon of als het goed weer is.
17	Je wilskracht, je passies, alles is weg.
18	Dat is...Ik weet ook niet wat er dan gebeurt waarom je dus wél een master heb gedaan, een hele analytische baan hebt en vervolgens niet kan bedenken dat je eerst je sloffen even uittrekt en dan je broek weer aan of zo.
19	Ik deed alles en alleen in huis, het klussen, de tuin alles. Dan denk ik, hoe heb ik het ooit kunnen doen?
20	Omdat ik ook heel graag wil dat het allemaal lukt, dat ik dan niet toe wil geven en dat ik eigenlijk beter even 10 minuten iets anders kan gaan doen. Ik wil gewoon zoals de rest geen pauzes nemen. Gewoon lekker door.
21	Je zoekt wel de gemakkelijke dingen, je wordt een beetje gemakzuchtig. Bijna lui voor sommige dingen, maar dan omdat het moet. In heel veel dingen.
22	En ik merk nu, ik voel het nu ook al hoor, dan ben ik erover aan het praten en met name als ik over de COVID aan het praten ben, daar gaat mijn ademhaling ook de verkeerde kant op of zo op de één of andere manier. Ja ik weet niet waarom, maar alsof het dan inderdaad teveel is.
23	Ja dan denk ik gewoon ook wat een heel leuk weekend, voel me superblij. Ja, ik heb ik voel me beroerd in fysiek, zeg maar, maar ik kan ook gewoon op bed liggen. Ik heb een fijn bed ik, er zijn verder weinig dingen die ik echt móet doen. Dus ik kan die rust nemen. Ja, ik heb inmiddels wel ook gewoon geaccepteerd dat het zo is
24	Nou ja, ik ben er ook wel heel actief mee bezig, hoor, om hè niet het glas halfleeg te zien, maar echt wel halfvol. Dus daar ben ik heel hard aan het werken om, hè, te genieten van de kleine dingen, wat ik wel heb.
25	Zo gezegd, ik ben een beetje moe omdat ik in het weekend naar een verjaardagsfeest ben geweest voor een paar uur en dat dreunt twee tot drie dagen door. Maar die ervaring heb ik en ik wist ook dat dat zou gebeuren. Maar daarom ben ik niet naar de verjaardag gegaan, ik moet toch ook gewoon een beetje aan het sociale leven deelnemen? Anders wordt het helemaal niets meer, hè?
26	En dat is waar je de uitzonderingen voor maakt, je maakt de uitzonderingen voor dingen waar je blij van wordt die vrolijke stofjes opleveren en dit is er een.
27	Dat heeft me. Ik denk wel twee jaar lang echt naar de ***** geholpen. Het is niet goed geweest voor mijn herstel. Want ja dingen moeten ook als ik me heel rot voel, dan sta ik op school kinderen weg te brengen. En als ik me nog rotter voel, dan sta ik gewoon te koken en als het echt niet gaat, dan sta je even goed knikkerbanen te bouwen met kinderen en spelletjes te doen. Want je moet wel. En mijn man die werkt. Dus een hoop komt ook op mij aan.
28	En het is ook, daar had ik het toevallig met nadat we dit aan het bespreken waren, met mijn man ook over van het is niet dat ik rust en uitrust. Ik moet rusten, want ja, ik kan gewoon niet de hele dag op zijn. Maar het is niet dat ik dan als ik op ben en ik ga even een half uurtje liggen dat ik daarna gewoon weer helemaal lekker in mijn energie zit.
29	Ik ga dus heel graag de natuur in en wandelen en dan zeg ik altijd, dat kan mijn accu weer opladen. Ik ga niet op de bank liggen om te slapen, dat lukt niet. En dat wil ik niet.
30	En ook na het weekend is het bijvoorbeeld dat ik wel echt maandag, dan plan ik ook nooit iets, want dan moet ik echt weer even rustig aan doen.
31	Ja nou ja en een hoop dingen opschrijven in mijn agenda ook, ook reminders voor mezelf. En vaak zet ik inderdaad ook een wekker. Maar vooral ook, hè dingen die mensen vertellen van, hé vriendin gaat volgende week op vakantie op skivakantie naar Oostenrijk en ze gaat daar en daar naartoe. Ja, dat dat schrijf ik op, want anders dan ja, dat ben ik gewoon weer kwijt.
32	Ja, dan probeer ik dus dat soort meetings voor te bereiden en dan merk ik dat ik gewoon heel erg veel in mijn hoofd zit. Maar dan ben ik nu dus gewoon in Notion alles aan het uitschrijven wat ik dan allemaal bedenk om gewoon een beetje orde te scheppen.

33	Dus, daar put ik heel veel energie uit en ja, door de dag heen af en toe bewust rust nemen wandelen dus op een heel langzaam tempo rustig je stapjes maken. Niet dat ik het fysiek niet aankan, maar omdat dat juist die rust en dat tempo ook in mijn hoofd rust geeft.
34	Ja maar toch onbewust ook, want je voelt het op dat moment niet zo omdat je bezig bent, maar dan op een gegeven moment, zegt hij van 'oh, wat ben je goed aan het sporten?' En dan denk ik 'hoeeee.!? Dat was ik niet, maar dankjewel.
35	Ja, ik vind het wel leuk. Ik vind het ook wel leuk dat er dan zo gesproken wordt. Dan voel je je wat minder eenzaam [lacht ongemakkelijk], het is ook wel gezellig.
36	Nou dat ik daar dus geen energie in hoeft te stoppen om me druk te maken over hoe dynamisch het verkeer op dat moment reageert, hè? Dus als het verkeer voor mij wordt afremt of het wordt wat druk en daardoor zakt de snelheid onder de gewenste snelheid van zeg maar honderd km per uur en we gaan daar naar negentig, tachtig, zeventig dan houdt mijn auto automatisch afstand. En ik hoef dus zelf minder aandacht en energie in dat autorijden te stoppen.
37	Nee, mijn horloge, die zakt gewoon alleen maar in de loop van de dag. Daar zit geen, ook al ga ik gewoon even liggen of doe ik oefeningen of nou, dan voel ik mezelf wel iets. Nou denk ik van nou, ik ben even tot rust gekomen. Echt geeft soms wel een blauw in de stress aan. Gewoon dat ik gewoon in de rust ben, maar die geeft geen oplaadmomentje aan, terwijl het voor mezelf wel als oplaadmomentje voelt.
38	Nee. Ik laat mijn leven niet door het horloge dicteren. Nee, het is voor mij is gewoon even kijken van hoe is die situatie? Ik accepteer niet dat ik zeg van je hebt goed geslapen, voel ik me toch moe, maar op dat moe zijn, dan doe ik weer iets anders. Ik wil niet moe zijn. Het is gewoon een beetje informatie wat je zegt het gaat niet zo slecht met je. Dus je kunt iets doen.
39	Of als je bijvoorbeeld ergens een soort hulp nodig hebt dat je een soort knopje kan invullen van oké, nou, wat is nu het probleem? Ik laat het even zien met mijn videotje en wat hij dan gewoon zegt, oké, ga dan nu zitten, trek je sokken uit of? In een stappenplan.
40	Als je erover nadenkt, voel je wel dat je moe wordt, maar je hebt het niet door, niet actief zeg maar. Ja, als er op dat moment technologie zou zijn die dat zou kunnen aanvoelen of weet ik veel en die zou ook kunnen ingrijpen. Van 'ik merk dat het even tijd is om even 5 minuten dit te doen', zeg maar.
41	Dus ik denk misschien wat Long COVID betreft, dat die Fitbit nog wel wat specifiek kan voor deze doelgroep. Niet gericht op sporten en goed je best doen maar gewoon meer op spanning voorkomen.

Appendix F: Flyer conclusions for participants

This section will include the conclusions that are specified for the participants and shared with them. Apart from the conclusions from this thesis, this will also include a list of findings regarding strategies and (technological) tools they can eventually implement in their lives. It should be noted that the interests of the participants differ from researchers and other interested people reading this thesis. Thus, communication with them on the conclusions is adjusted to what the first researcher believed to be valuable for their process and participation. Besides, their needs and desires to process information are (partially) determined by their Long COVID, so the document is adjusted to enable reading it.

Long COVID in het dagelijks leven:

Onderzoeksresultaten en inzichten van andere deelnemers

Allereerst wil ik je hartelijk bedanken voor je deelname aan dit onderzoek. Jouw ervaringen en die van andere deelnemers zijn heel belangrijk geweest om beter te begrijpen hoe Long COVID je dagelijks leven beïnvloedt en hoe technologie daarin een rol speelt.

Universele bevindingen

De resultaten van het onderzoek laten zien dat Long COVID vrijwel alle aspecten van het dagelijks leven beïnvloedt. Velen van jullie zijn actief op zoek naar manieren om met deze klachten om te gaan. We zien dat de huidige hulpmiddelen vaak gericht zijn op lichamelijke klachten, terwijl juist de cognitieve functies – zoals concentratie en geheugen – zo belangrijk zijn in het dagelijks leven. Daarom wilden we in dit onderzoek beter begrijpen hoe technologie kan ondersteunen bij de cognitieve uitdagingen die in het dagelijks leven ontstaan door Long COVID. We hebben gekeken naar jullie ervaringen met strategieën en hulpmiddelen, waaronder technologie.

Resultaten in het kort:

 **Long COVID raakt alle aspecten van het dagelijks leven.** Dit houdt, o.a., in moeite met dagelijkse activiteiten, sociale contacten, mentaal welzijn en het beheersen van klachten. Het is een voortdurende zoektocht naar balans tussen klachten, verwachtingen en persoonlijke wensen.

 **Mensen ontwikkelen zelf oplossingen om met hun klachten om te gaan.** Veel van jullie ontwikkelen zelf manieren en trucjes om met de klachten om te gaan in het dagelijks leven.

 **Technologie helpt al, vooral als een soort zelfhulpstrategie.** Op dit moment gebruiken jullie technologie vaak om meer inzicht te krijgen in jullie klachten en om ze beter te beheersen. Het ondersteunt ook bij bepaalde taken die veel denkwerk vragen.

 **Technologie kan nog beter.** Jullie gaven aan dat technologie al helpt, maar dat het nog beter zou kunnen aansluiten bij de specifieke behoeften van mensen met Long COVID.

 **De toekomst van technologie ligt in het ondersteunen van informatieverwerking en het beter managen van klachten.** Jullie zien potentie in technologie die vooral ligt in de ondersteuning bij informatieverwerking en het optimaliseren van technologie die helpt om jullie klachten te managen.

Inzichten van andere deelnemers:

In dit deel bespreken we kort methodes die andere deelnemers in dit onderzoek hebben gebruikt om hun dagelijks leven met Long COVID makkelijker te maken. Misschien heb je er zelf ook iets aan. Het is goed om te weten dat dit een momentopname is en dat dit onderzoek een eerste verkenning was. De lijst is dus niet compleet of af.

 **Omgaan met mentale uitdagingen:** De meeste deelnemers gebruiken manieren om mentaal sterker te worden en zich beter te voelen.

 **Trucs voor dagelijkse activiteiten:** Veel deelnemers passen activiteiten aan of bedenken trucs om ze toch te kunnen doen.

- Dit gaat vaak over het makkelijker maken van taken, minder stress ervaren en grenzen stellen. Een veelgebruikte manier is om activiteiten eenvoudig te houden. Maar mensen gebruiken ook andere trucs om bijvoorbeeld koken of afspreken met mensen beter te laten passen bij wat ze aankunnen.
- Een specifieke truc is het gebruiken van **lijstjes of notities**. Veel deelnemers gebruiken dit om hun geheugen te ondersteunen. Het helpt hen te herinneren aan dagelijkse dingen en afspraken, zoals wat ze moeten doen, wat er in een gesprek is gezegd of wat ze moeten kopen.

 **Manieren om te herstellen:** Veel deelnemers gaven aan dat ze actief bezig zijn met herstelmethodes.

- Dit kan betekenen dat ze hun hersenfuncties trainen, maar ook dat ze behandelingen zoals mind-body oefeningen gebruiken, bepaalde medicijnen nemen of meer de natuur opzoeken en wandelen.

 **Gebruik van technologie:** Bijna alle deelnemers gebruiken hulpmiddelen, vaak technologie, om met hun Long COVID om te gaan.

- Technologie wordt vooral gebruikt om meer inzicht te krijgen over hun gesteldheid. Een voorbeeld is een smartwatch die laat zien hoe zwaar een inspanning is.
- Ook worden hulpmiddelen zonder technologie gebruikt om te voorkomen dat ze overprikkeld raken. Hierbij moet je bijvoorbeeld denken aan oordoppen om geluid te dempen.
- Soms hebben mensen ook gemerkt dat technologie kan helpen bij het herstel. Dit kan bijvoorbeeld gaan om het controleren van andere gezondheidsproblemen via technologie, wat dan weer helpt om zich in het algemeen beter te voelen.

 **Technologie als ondersteuning in het dagelijks leven:** Voor sommige deelnemers helpt technologie om mee te doen in de maatschappij.

- Voor sommigen is technologie heel belangrijk om contact te houden met de buitenwereld. Het kan helpen om te praten met anderen of om mentale steun te krijgen van technologie (bijvoorbeeld via een Google Home).
- Technologie helpt ook om dingen te doen door de lichamelijke of mentale kant te ondersteunen. Veel deelnemers gebruiken technologie voor taken die veel denkwerk kosten, zoals informatie verwerken of koken (bijvoorbeeld via een app met recepten). Een voorbeeld voor autorijden is adaptieve cruise control, die helpt bij het concentreren.

 **Let op:** Technologie kan soms ook belemmerend zijn.

- Dit gebeurt vooral als de technologie niet goed past bij Long COVID, bijvoorbeeld een smartwatch die op sporters is ingesteld. Ook problemen met energie, veel telefoontijd en errors in technologie kunnen storend zijn.

 **Behoeftte aan meer ondersteuning via technologie:** Sommige deelnemers zouden graag meer ondersteuning van technologie willen om beter met hun Long COVID om te gaan.

- Ze denken dat technologie kan helpen bij taken die veel denkwerk kosten of bij het omgaan met klachten.
- Sommige deelnemers vinden dat de huidige technologieën verbeterd kunnen worden door ze aan te passen aan de behoeften van mensen met Long COVID.